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Helmholtz-Zentrum hereon

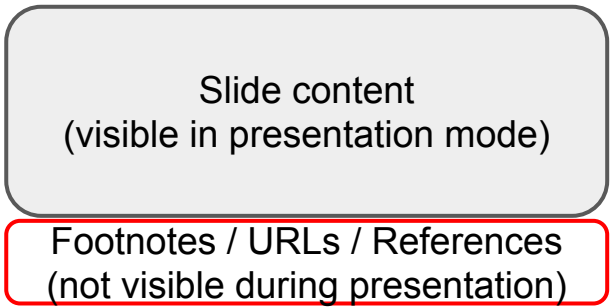
Understanding Transformers
Overview, Internals and Insights

November 2025

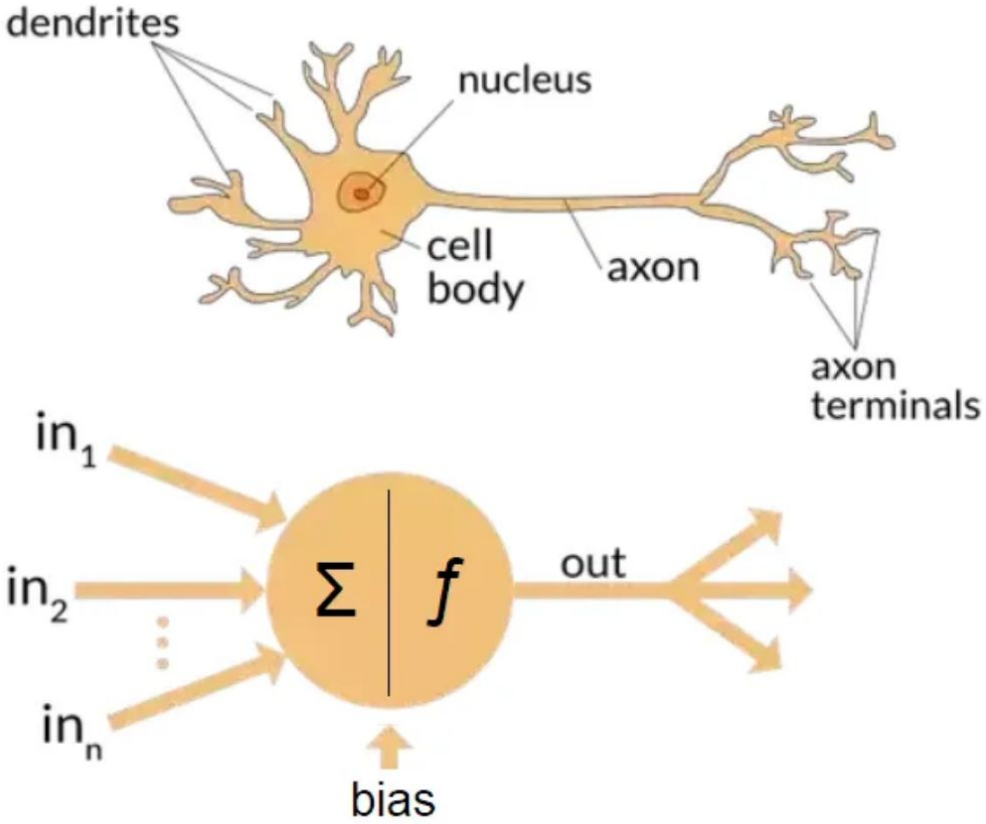
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Background notes / Prologue

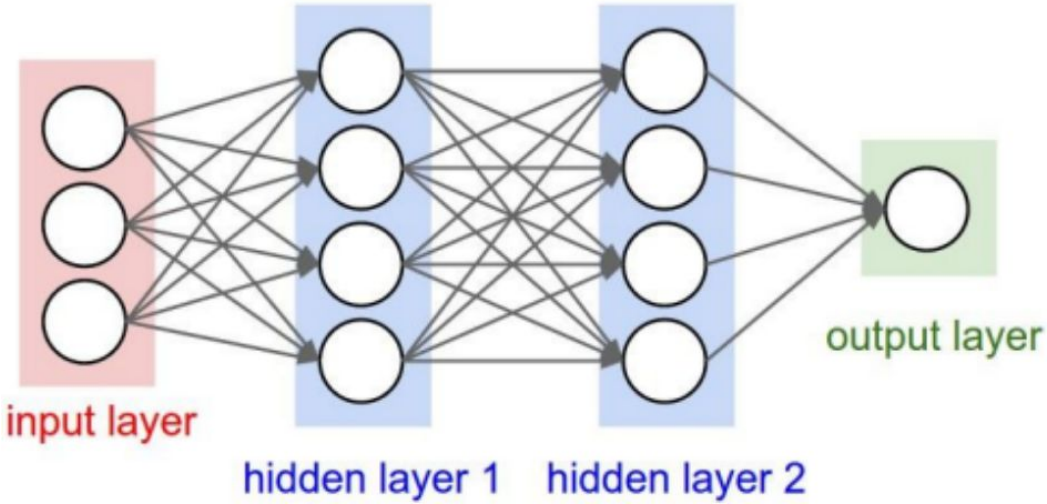
- This resource is intended to introduce neural networks, generative AI at a high abstract level and subsequently focus on **explaining Transformers** in more depth to a scientific audience.
- Transformers will be presented in the broader context of AI, building the stage from zero, step by step.
- We **will act** therefore **as a synthesizer** of many resources created by the broader AI community.
- Hence, many thanks! to all the creators of the helper material. **All credits** and references are visible on the **last slide**.
- Same references are also given in the **footnotes section** of each individual slide, which is **not visible in presentation mode**, but will be useful for referencing at home for extra study, if need be.



Artificial Neural Nets and Brain Parallels

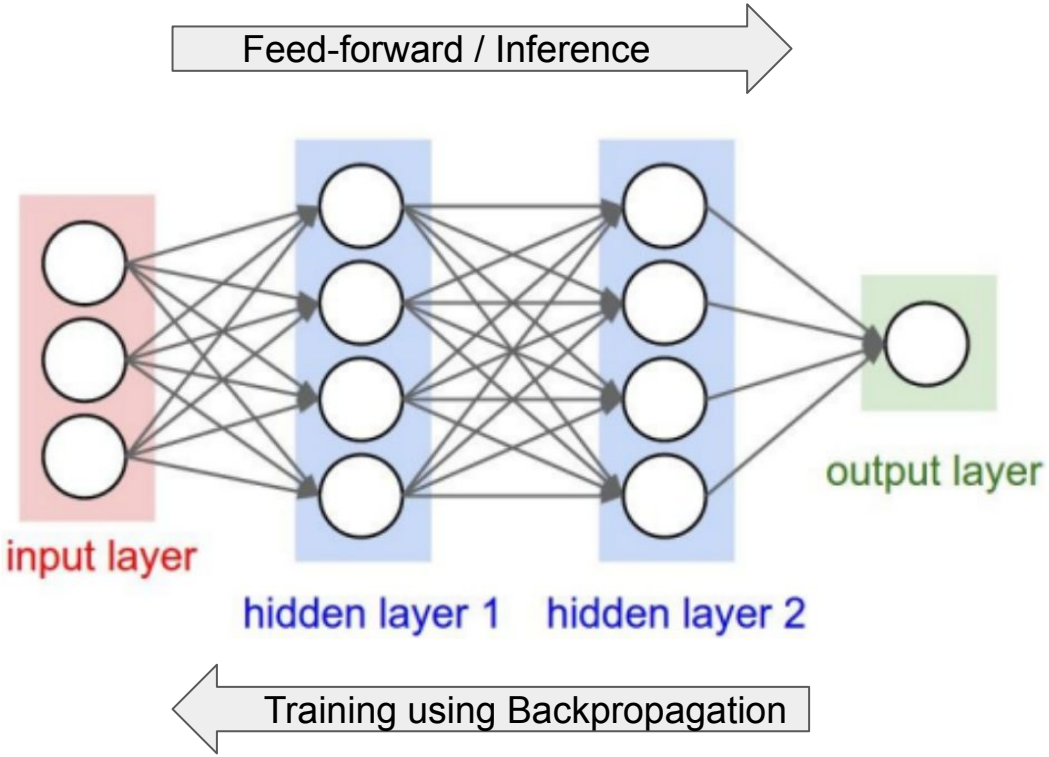


Artificial Neural Networks (ANNs)



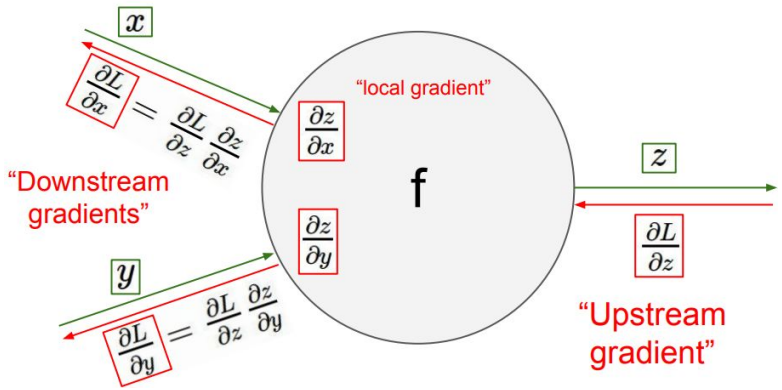
This is a **Multi-Layer Perceptron / MLP** and it is used as one of the building blocks of the Transformer.

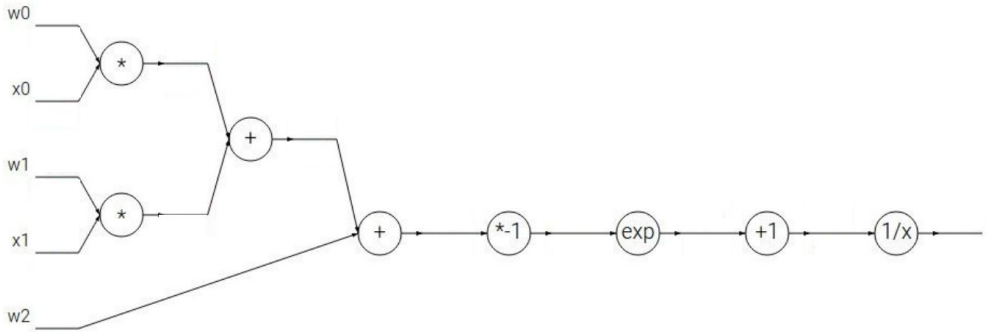
Backpropagation



CS231n course from Stanford University

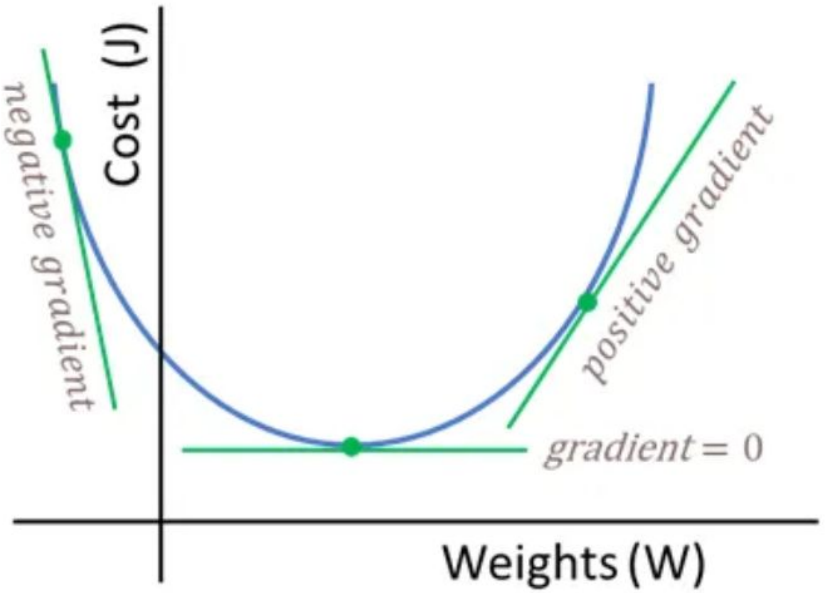
Backpropagation using Local Gradients and the Chain Rule





- The Chain Rule is implemented with the help of **local gradients**.
- We **recursively multiply** the local derivatives.
- Backpropagation is a **recursive** application of the chain rule, backwards through the **computational graph**.

Cost function



- Naming conventions in the literature: **Cost / Loss / Error** function

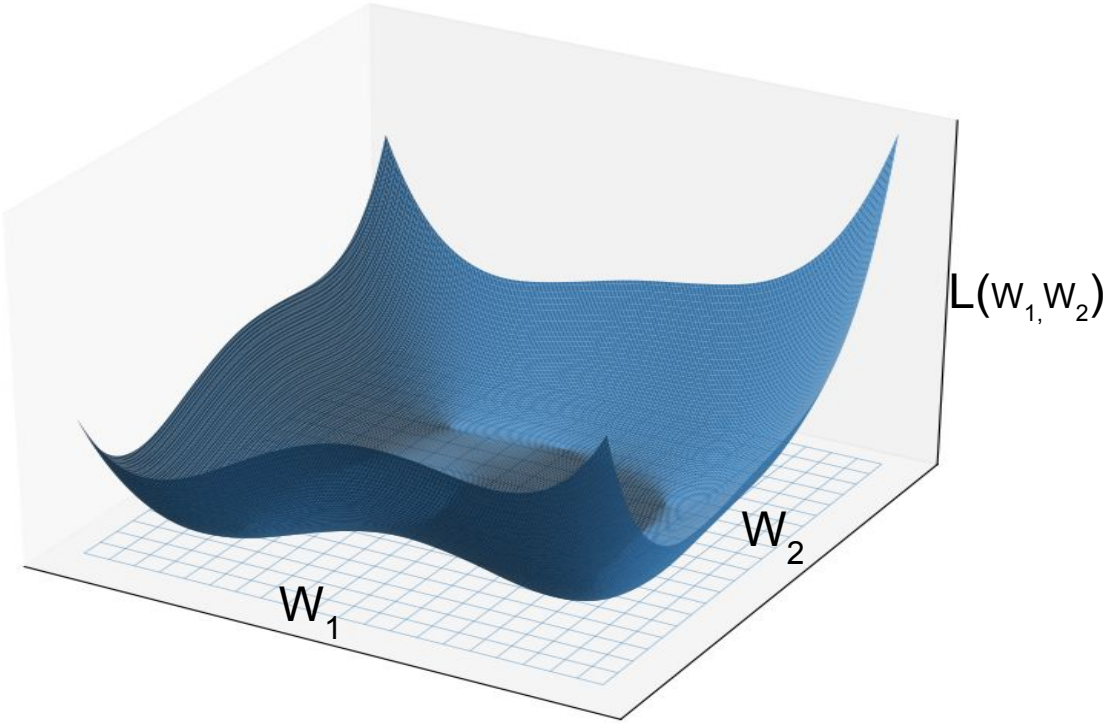
- The task is to find a **global minimum**, or **good** enough **suboptimal minima** points via the **gradient descent** optimization procedure, i.e. parameter search.

Weights update

$$W_{new} = W_{old} - \alpha \underbrace{\frac{dJ}{dW}}_{\text{gradient}}$$

- **J** and **L** are usual notations for the **Loss** / **Error** / **Cost** function, i.e. the difference between what the model **predicts** and what it should predict according to the **ground truth**.
- The **weights are updated** in the direction of **negative** gradient, so that the **cost function is minimized** as much as possible.

Scaling to Hyperspaces

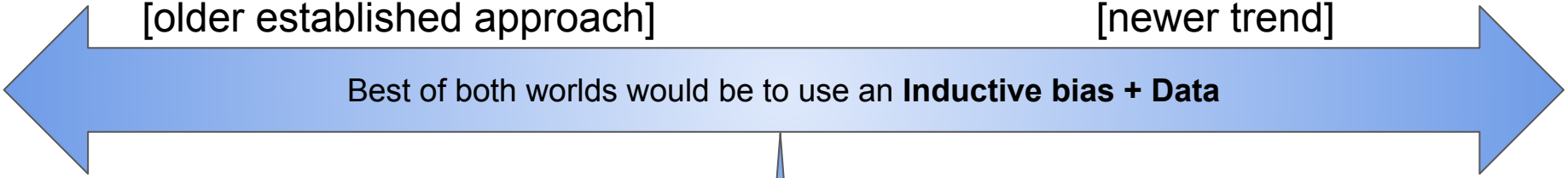


- The **cost function is** much more **complex** for a **typical neural net** => Millions and even billions of weights!
- Neural Network gradients have the ability to **navigate the many dimensions skillfully** though.
- And actually, having many dimensions **helps the gradient not to get stuck** in bad suboptimal points.

Modeling Spectrum in the current AI era

Discriminative models
[older established approach]

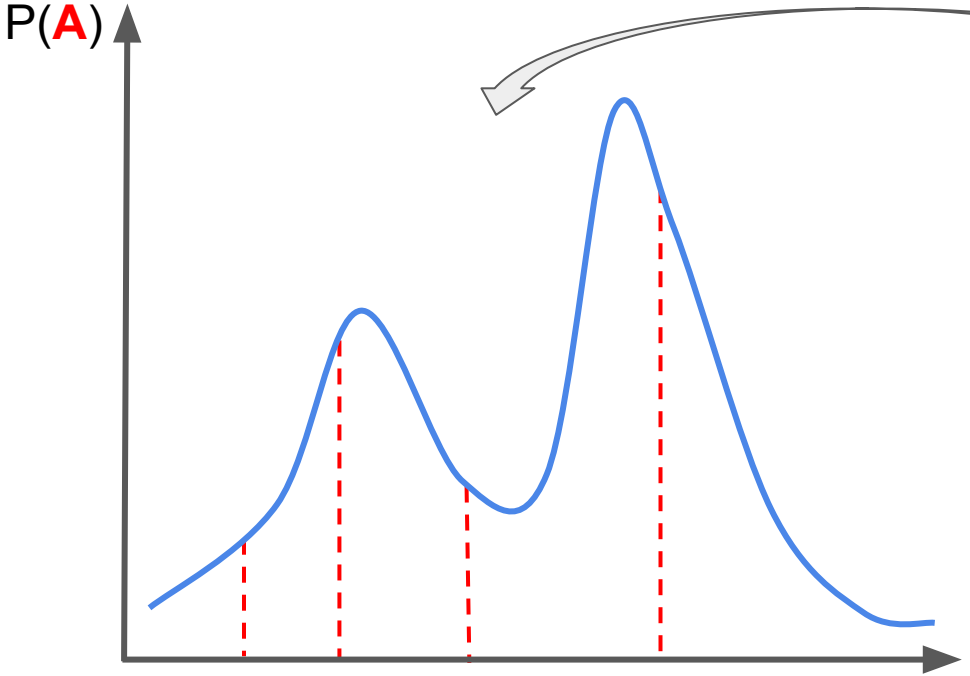
Generative models
[newer trend]



- Use **a lot of training data** / throw a lot of data at the problem task and **let the network "figure it out".**
- Predictions are limited to the training data domain, i.e. tied to the train dataset statistics, hence **poor out of distribution predictions.**

- Are capable of **out of domain generalization.**
- Lots of inductive bias, **a priori knowledge** is injected / enforced as a guide during training.
- **More creative** than discriminative models.

Encode a “**world model**” in the form of a very-high-dimensional / complex **probability distribution.**



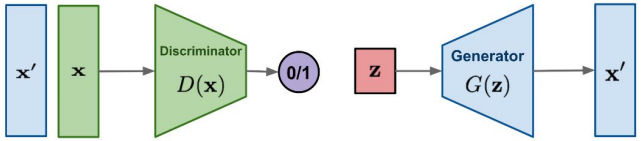


- Internet
- Books
- Images
- Videos
- Audio
- Robotic-Arm configurations
- Etc.

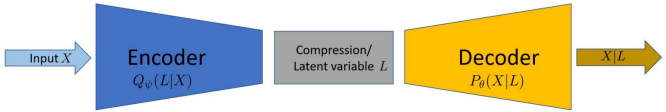
User prompts will statistically “sample **A**nswers” from the learned model.

Mainstream Generative AI models

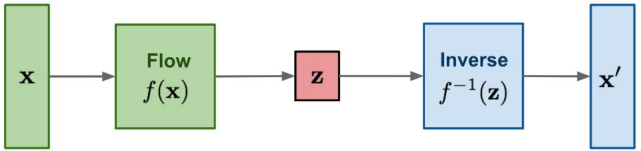
- Generative Adversarial Networks / GANs -> **Adversarial** training / **Arms race**



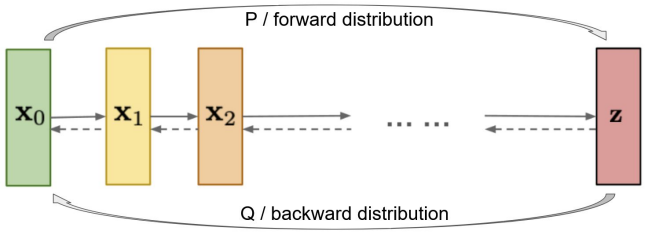
- Variational Autoencoders / VAEs -> Autoencoders, but with **latent space**



- Flow-based models -> **Invertible mapping** between distributions



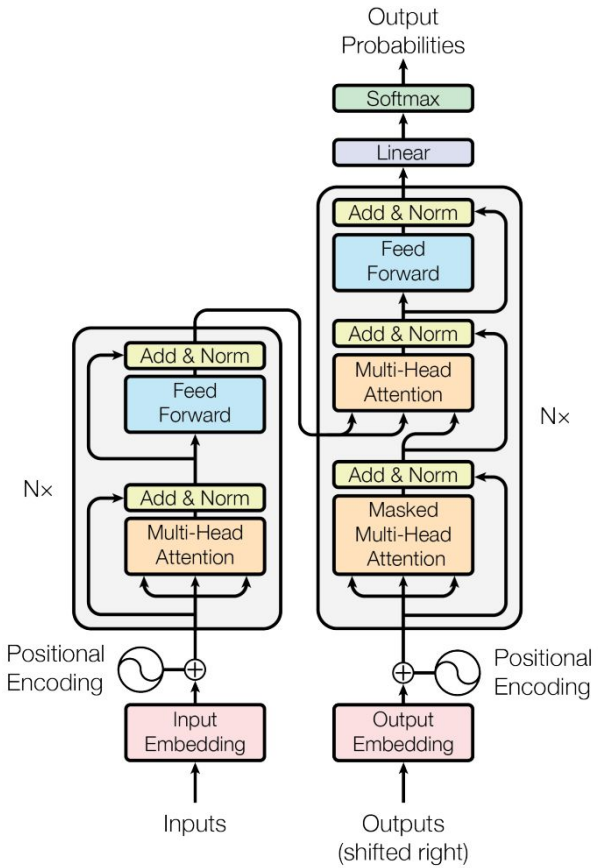
- Diffusion models -> **Markov chain** for **denoising** intermediate states



- **Transformers -> we will focus our attention on them**

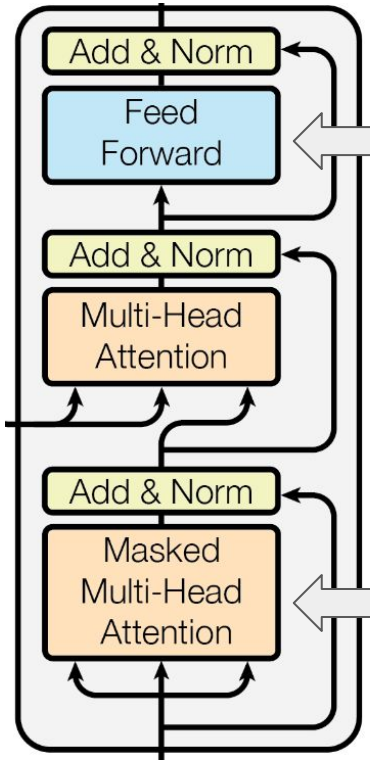
Coming Soon

Transformers



“Attention is all you need” paper from 2017 by Vaswani et al.

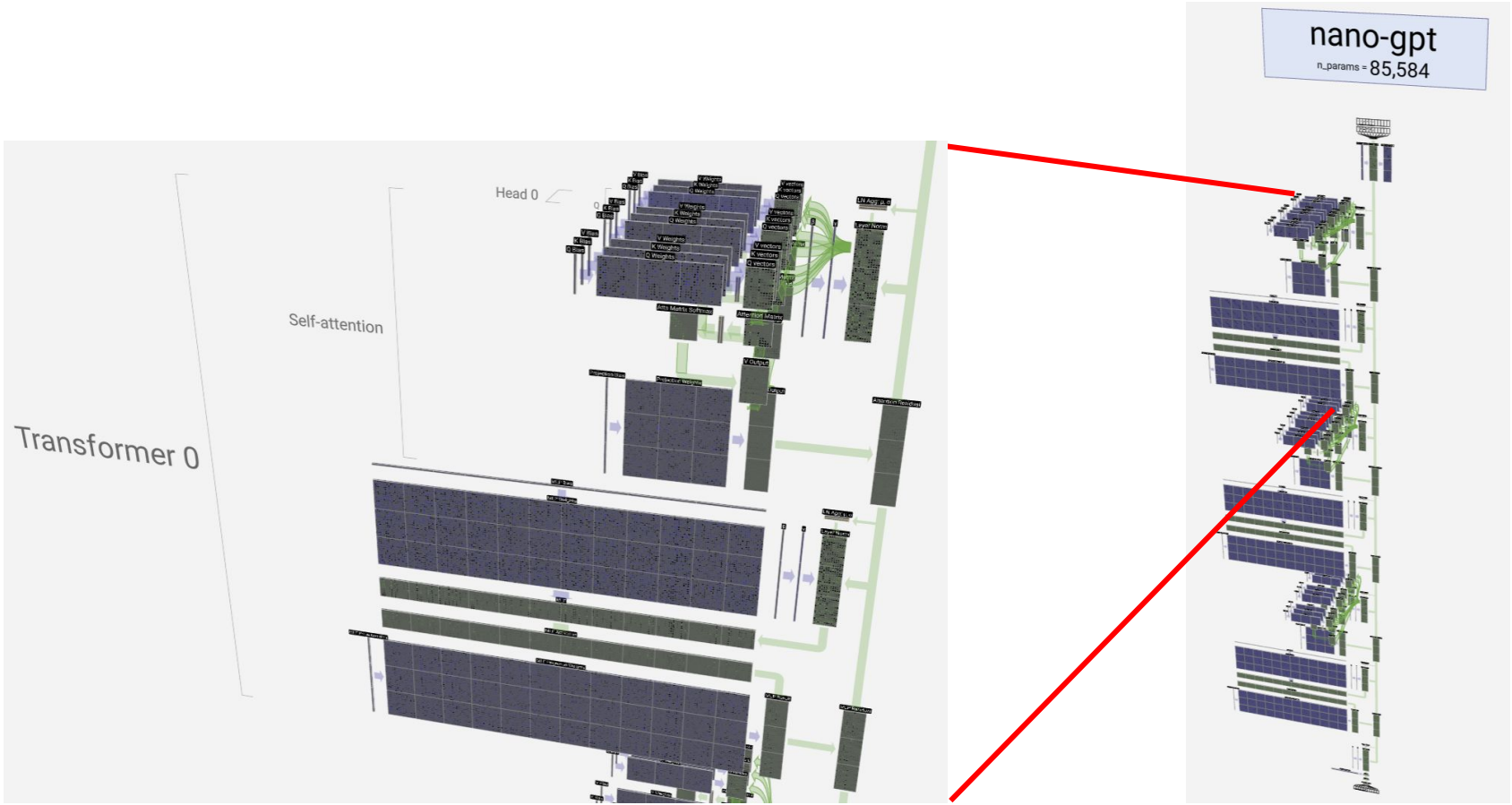
Conceptual Transformer Block



Secondly, a **Multilayer Perceptron (MLP)** crunches the data that was communicated among tokens in the previous self-attention phase.

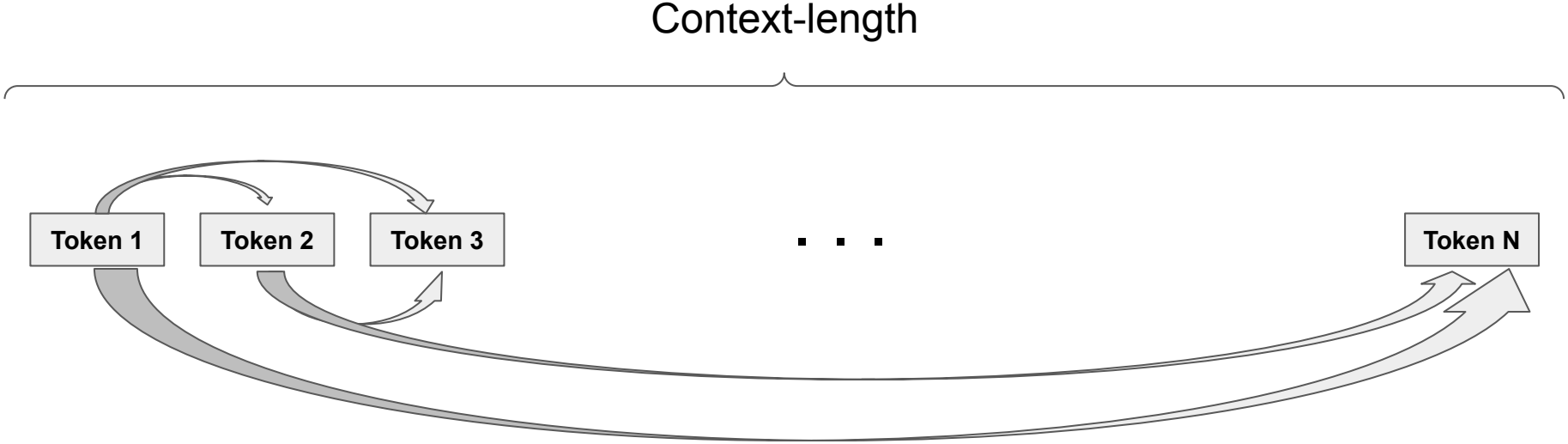
First, the tokens **communicate** between themselves / they attend to each other (**self-attention**).

Realistic GPT Block



*Visualize the physical structure of an LLM interactively at -> **https://bbycroft.net/llm***

Self-Attention



➢ **All tokens communicate with one another**.

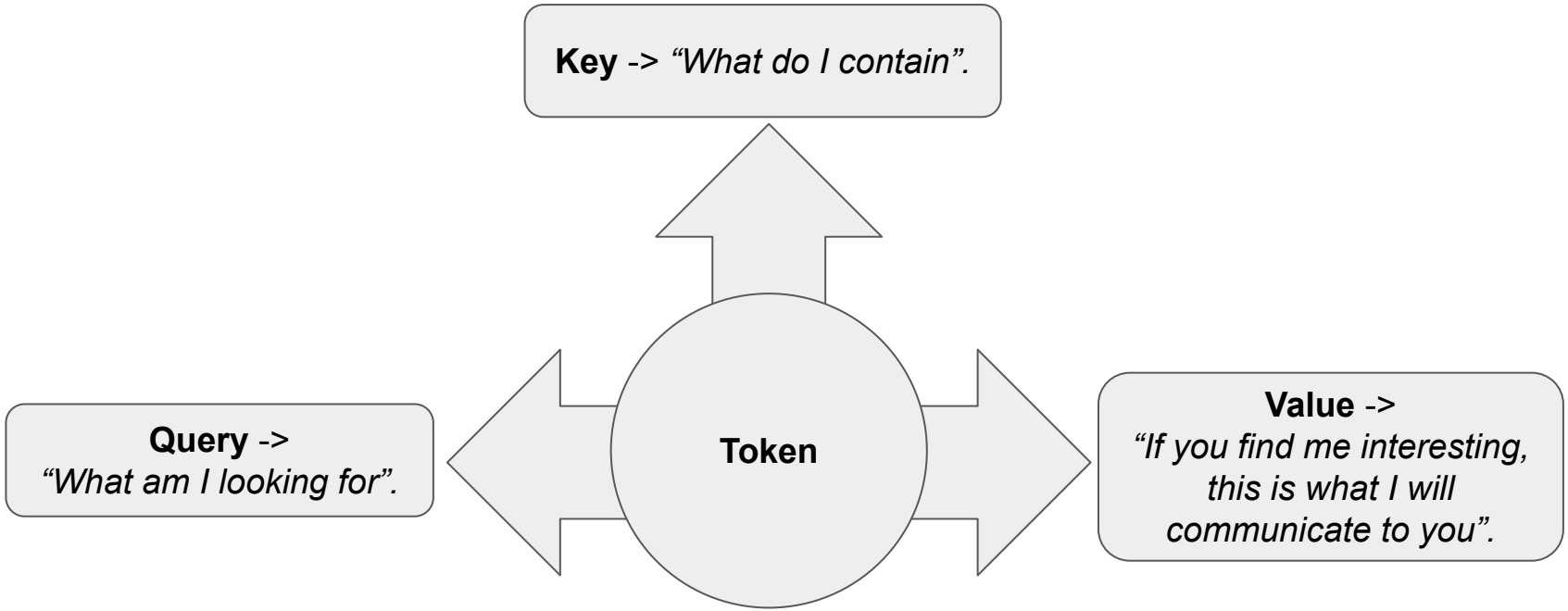
➢ This is computationally expensive *[i.e. $O(N^2)$ complexity]* because **each token has to look at every other token** to compute an **attention score / attention weight**.

Increasing the Context Length

- **Sparse Attention** Mechanisms
- Linformer
- Reformer
- Longformer
- **Blockwise** Attention
- Ring Attention

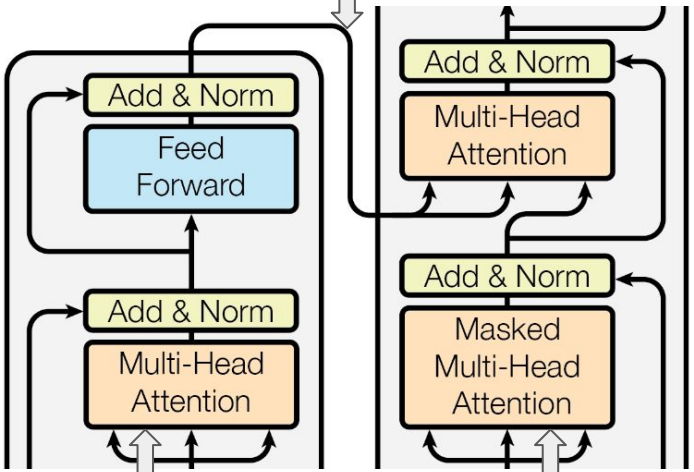
Reducing attention complexity is a very hot topic today🔥. This would allow for **more context memory** to be used when tackling problems. More memory means **longer chains of thinking => thinking is computationally distributed over more tokens**, which generally helps producing more accurate answers and can **reduce hallucinations** .

Key, Query and Value Embeddings



Self-Attention vs Cross-Attention

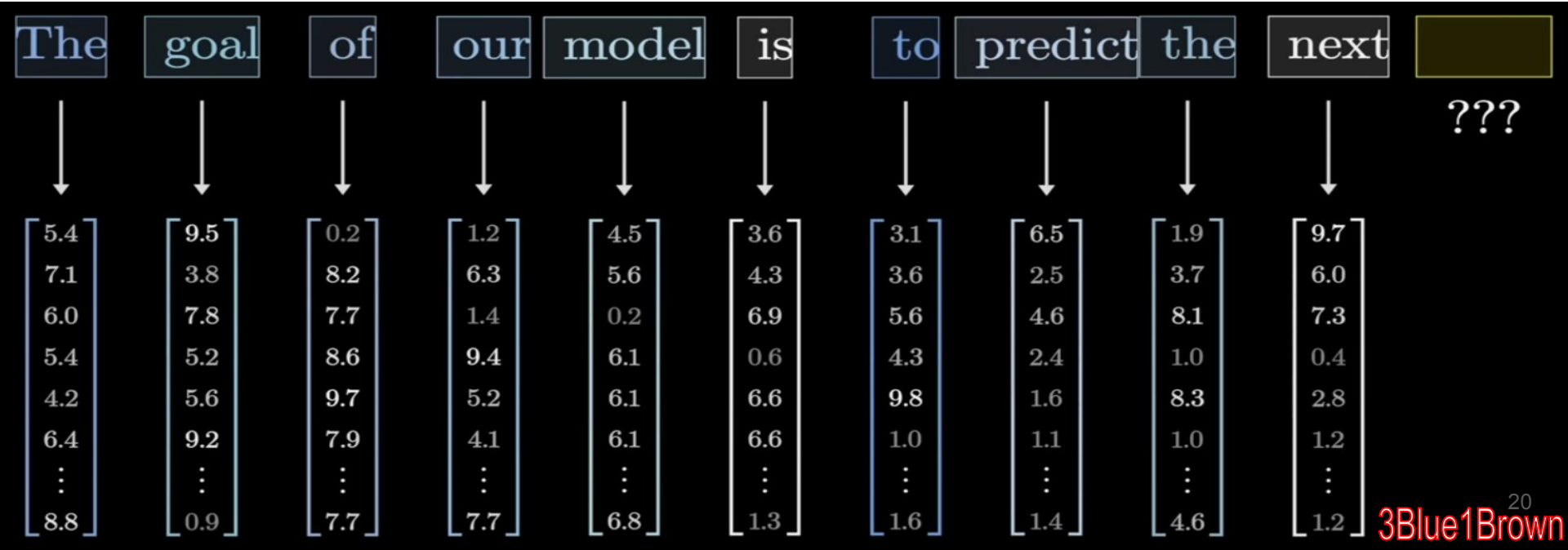
Cross-Attention -> the *queries* come from the **decoder**, whereas the *keys* and *values* are from the **encoder** side.



Self-attention -> the *key*, *query* and *value* vectors are related to the **same entity**, either the encoder, or the decoder.

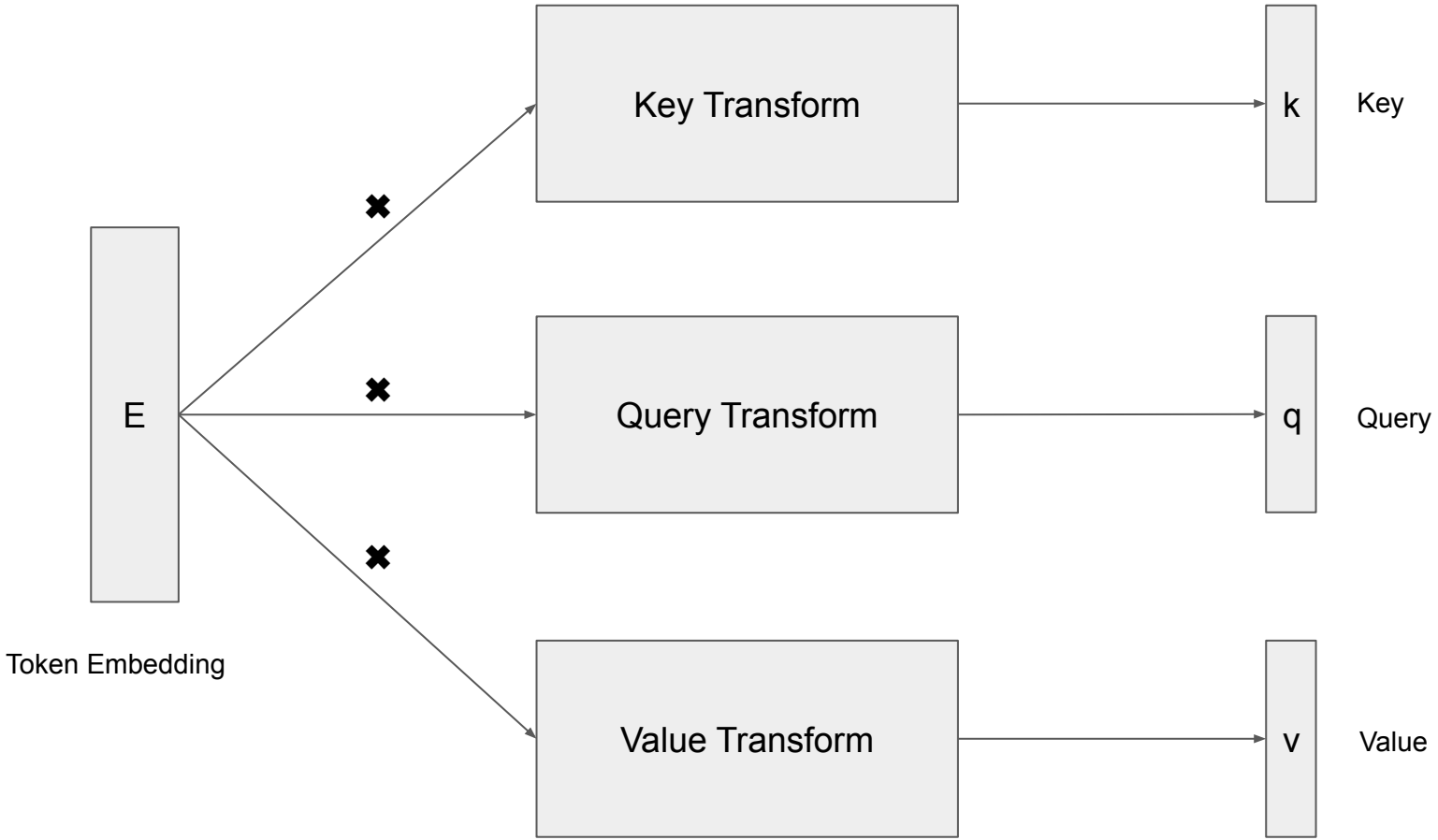
Token Embeddings

Tokens become **positional vectors in a semantic space** and are stored in an "**Embedding Matrix**". Our task will be to predict the next token, i.e. the embedding vector of the next token.



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Transformation Pipelines



Transformation Matrices

$$\mathbf{E} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_{\text{emb_dim}} \end{bmatrix}$$

$$\text{Key Matrix } \mathbf{W}_K = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1\,d_k} \\ w_{21} & w_{22} & \cdots & w_{2\,d_k} \\ \vdots & \vdots & \ddots & \vdots \\ w_{\text{emb_dim},1} & w_{\text{emb_dim},2} & \cdots & w_{\text{emb_dim},d_k} \end{bmatrix} \quad \mathbf{W}_K \in R^{\text{emb_dim} \times d_k} \qquad \mathbf{k} = \mathbf{E}\,\mathbf{W}_K$$

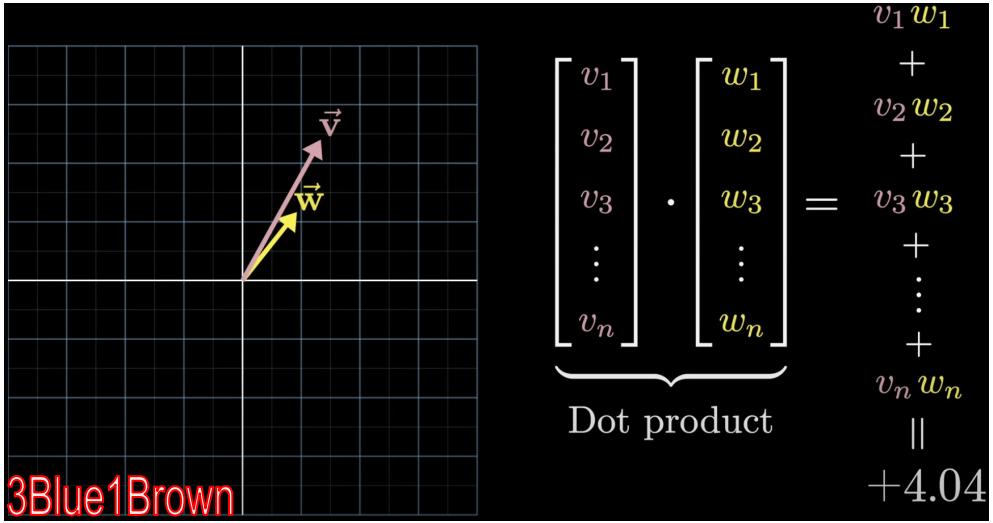
$$\text{Query Matrix } \mathbf{W}_Q = \begin{bmatrix} w'_{11} & w'_{12} & \cdots & w'_{1\,d_q} \\ w'_{21} & w'_{22} & \cdots & w'_{2\,d_q} \\ \vdots & \vdots & \ddots & \vdots \\ w'_{\text{emb_dim},1} & w'_{\text{emb_dim},2} & \cdots & w'_{\text{emb_dim},d_q} \end{bmatrix} \quad \mathbf{W}_Q \in R^{\text{emb_dim} \times d_q} \qquad \mathbf{q} = \mathbf{E}\,\mathbf{W}_Q$$

$$\text{Value Matrix } \mathbf{W}_V = \begin{bmatrix} \tilde{w}_{11} & \tilde{w}_{12} & \cdots & \tilde{w}_{1\,d_v} \\ \tilde{w}_{21} & \tilde{w}_{22} & \cdots & \tilde{w}_{2\,d_v} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{w}_{\text{emb_dim},1} & \tilde{w}_{\text{emb_dim},2} & \cdots & \tilde{w}_{\text{emb_dim},d_v} \end{bmatrix} \quad \mathbf{W}_V \in R^{\text{emb_dim} \times d_v} \qquad \mathbf{v} = \mathbf{E}\,\mathbf{W}_V$$

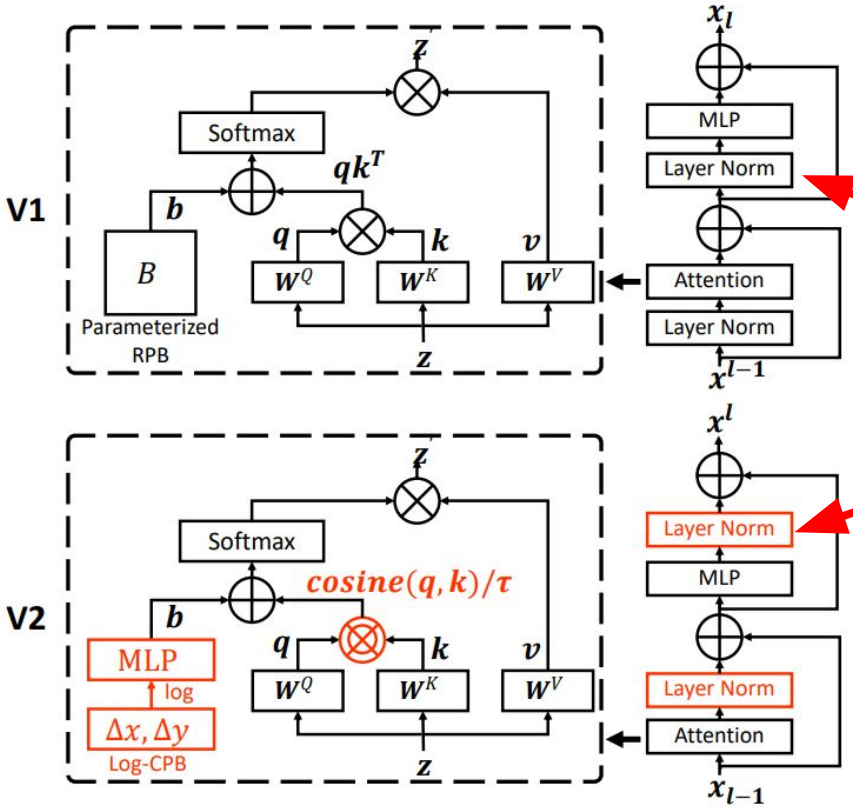
An embedding **transform** is a **multiplication** with a specialized **matrix**, the parameters of which the transformer needs to **learn**.

Mathematically Expressing Self-Attention

- The dot product **Query ▪ Key** is the attention score, where **Query** and **Key** are **embedding vectors** derived from the token embedding via matrix transformations.
- Dot product measures **similarity between vectors** => **Attention** can be interpreted as the **alignment** between the **Key** and the **Query** vectors (i.e. two tokens find each other interesting).
- Instead of the dot product, other measures can be used, like the **cosine similarity** for example (see **Swin Transformer Version 2** paper).



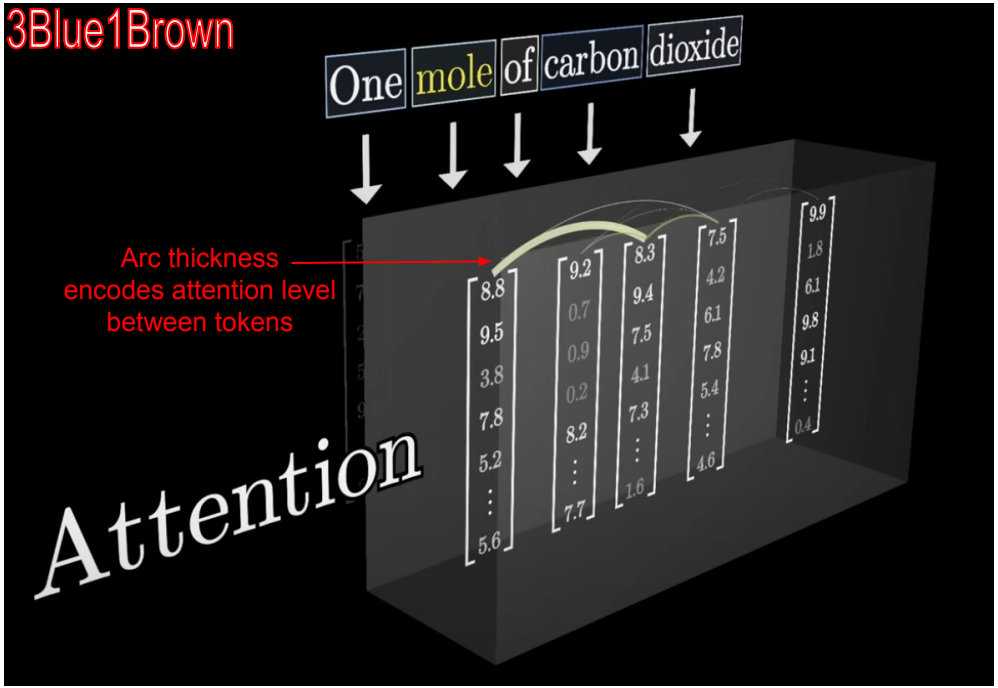
Cosine similarity



Notice the **pre**-Layer Norm vs **post**-Layer Norm switch.

"Swin Transformer V2: Scaling Up Capacity and Resolution" by Liu et al.

Discerning meaning among different contexts

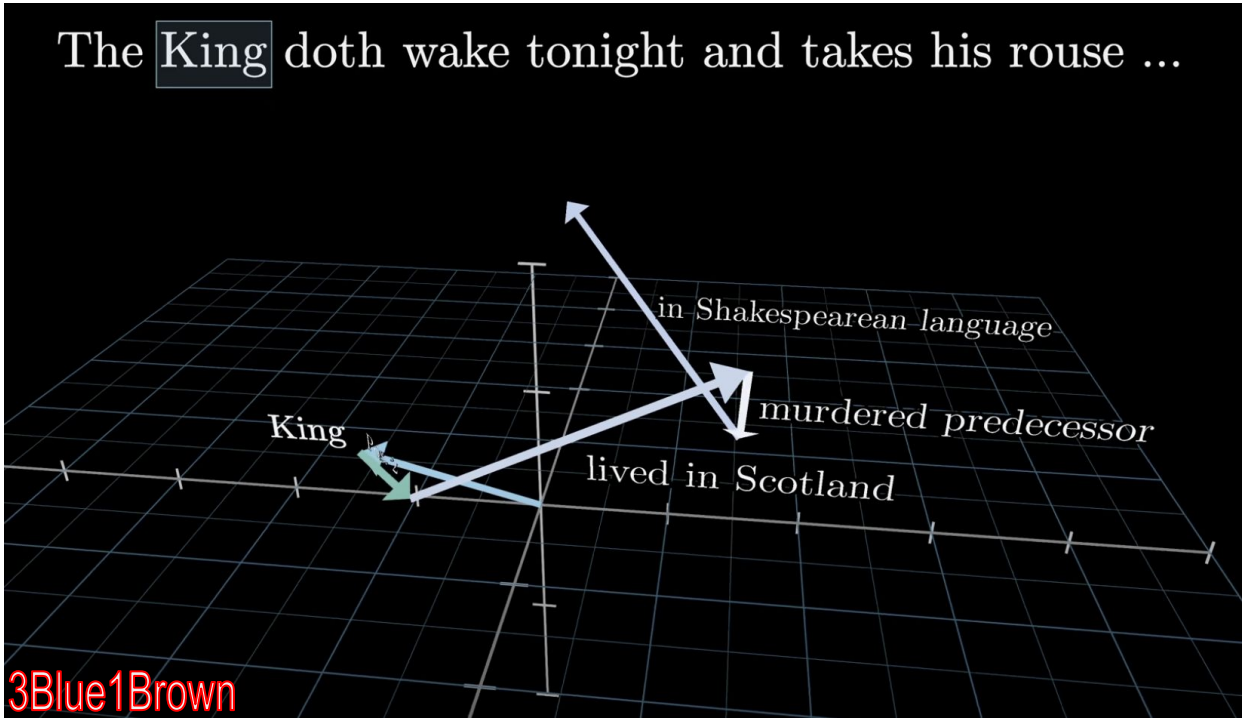


The **mole** on his face was benign.

The **mole** lives underground.

Attention will capture the meaning of a word **from the surrounding context**, using **weighted averages of neighboring embeddings**.

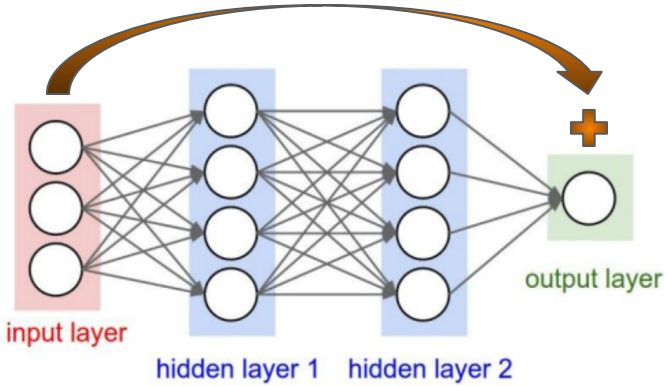
Navigating the Semantic Space via Attention



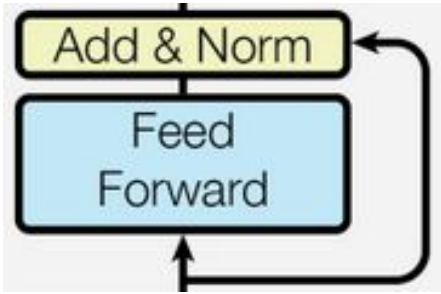
The embedding of the word “King” **absorbs via attention** the **embeddings from the other words** in Shakespeare's work, **as well as of the questions and statements that the users make** in their conversation with the Transformer model.

Computation Phase / Feed-Forward MLP

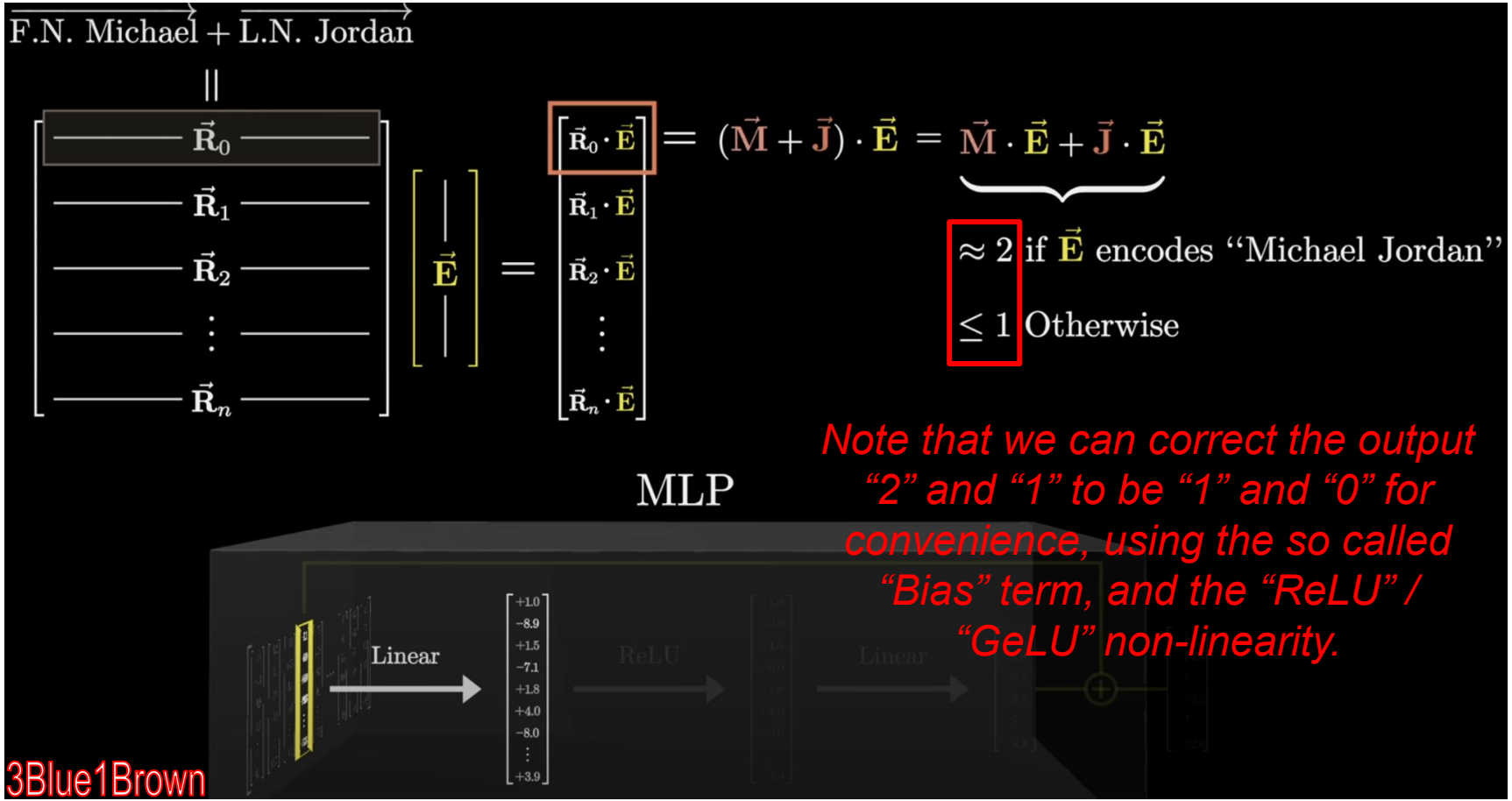
- After the communication between tokens is finished, an MLP has to **“think on what was said”** during the self-attention phase.
- This basically means that **new features are derived** after the communication phase **using a database of facts** encoded by the MLP.
- **Skip connections** can be viewed as **enriching the embedding**, but also helping the learning process from a technical perspective.
- **Normalization** is also very important for **stabilizing learning** and fostering convergence.



Let’s fast forward back to what we have already seen, **and notice! the additional skip connection.**

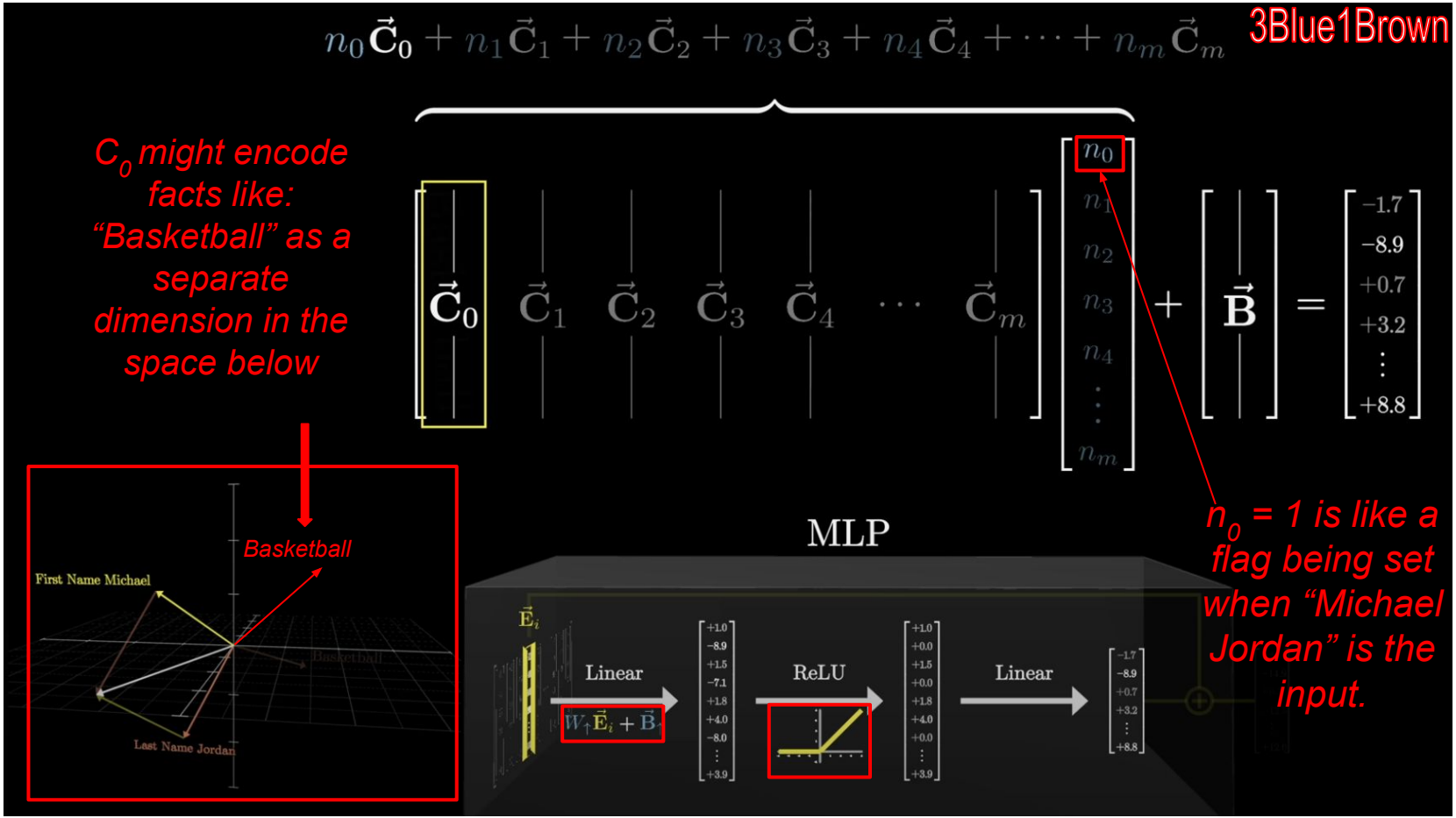


MLP Database of Facts



Note that we can correct the output "2" and "1" to be "1" and "0" for convenience, using the so called "Bias" term, and the "ReLU" / "GeLU" non-linearity.

MLP Database of Facts, Layer 2



“Grandmother cells” analogy in Neuroscience

Jennifer Aniston strikes a nerve

Single brain cells show selective response to specific celebrity photos.

Roxanne Khamsi

Is a single cell in your brain devoted to Jennifer Aniston or Bill Clinton? Maybe so, according to new research.

A recent experiment showed that single neurons in people's brains react to the faces of specific people. Researchers see the findings as evidence that our brains use fewer cells to decode a given image than previously thought.

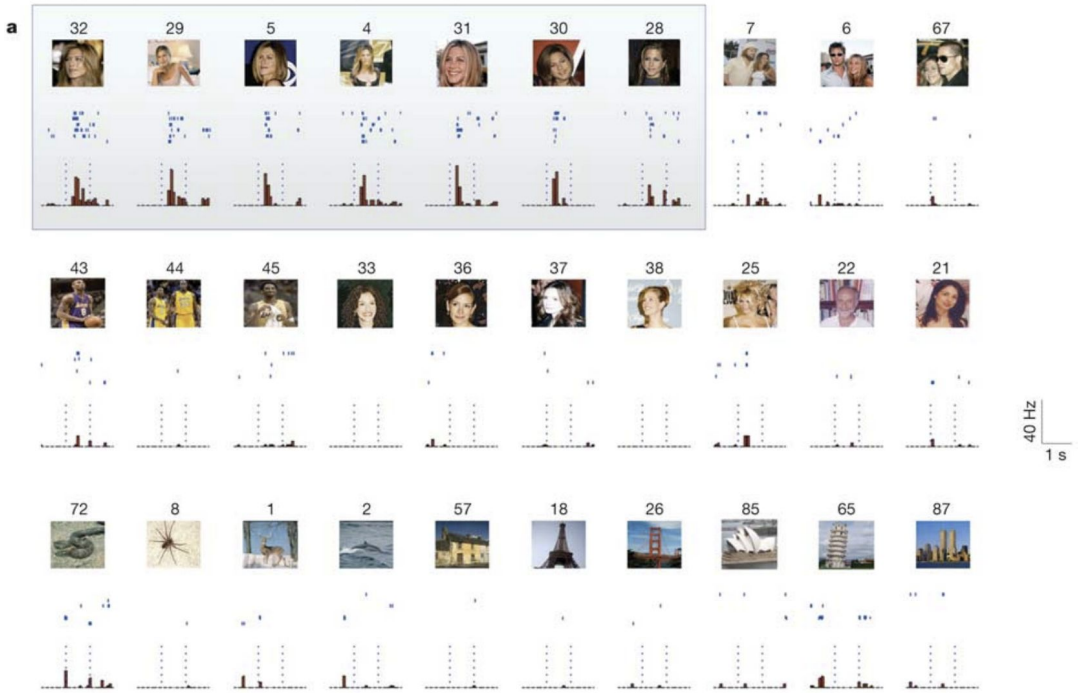
The subject of visual processing has sparked much scientific speculation in the past. Exactly how our brains extract meaning from an image remains unclear. At one end of the spectrum of possibilities, a network of cells would process various bits of information in a scene and piece it all together to form an understandable picture.



Snaps of Aniston made a single neuron in one subject’s brain light up

© *Photo/Lionel Cironneau*

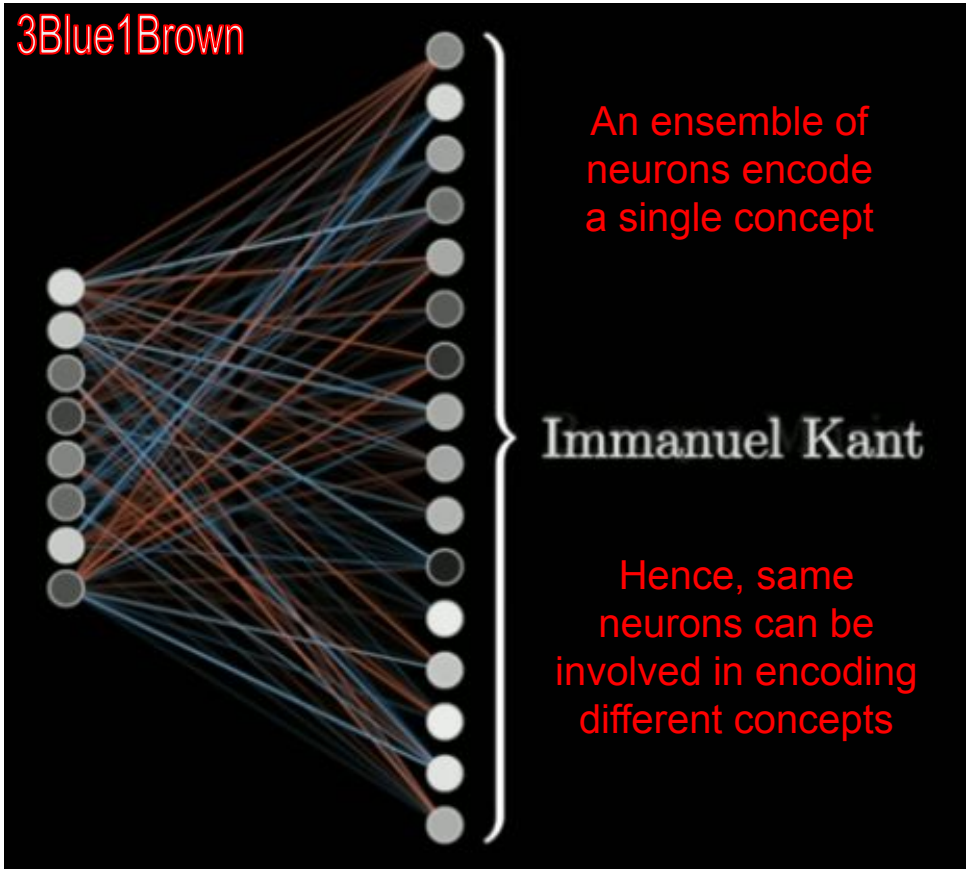
nature.com/news



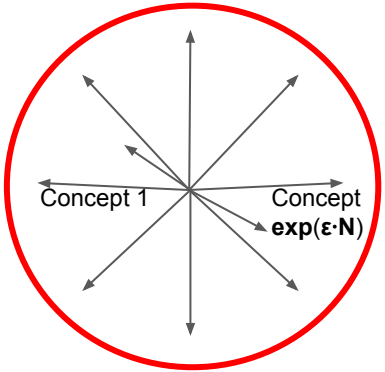
“**Invariant visual representation by single neurons in the human brain**”
R. Quian Quiroga et al., Nature 2005

“This unit was selectively activated by pictures of the actress Halle Berry as well as by a drawing of her”
“Notably, the unit was selectively activated by the letter string ‘Halle Berry’ ”

Superposition

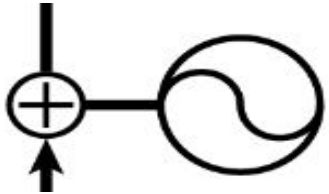


A much more usual scenario is that an **ensemble of neurons encodes a concept**, rather than a single "grandmother cell". This leads to the principle of **superposition**, in which a lot of the **same neurons will be involved in encoding several concepts**. It's a **combinatorics game for semantics** that becomes **very expressive when we have very high dimensional spaces** like the 12,288 embedding size in the case of GPT-3 175B (Davinci) for example.



N-dimensions

Positional encoding



Positional Encoding

$$\underbrace{\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_d \end{bmatrix}}_{\text{Content embedding } \mathbf{E}_{\text{token}}} + \underbrace{\begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_d \end{bmatrix}}_{\text{Position embedding } \mathbf{E}_{\text{pos}}} = \underbrace{\begin{bmatrix} e_1 + p_1 \\ e_2 + p_2 \\ \vdots \\ e_d + p_d \end{bmatrix}}_{\text{Final input embedding } \mathbf{X}}$$

- The transformer treats the tokens as a **Bag of Words (BoW)**.
- We need to give each token a label that specifies its position in the form of a **counter ID**, for instance.
- It's noteworthy that the positional encoding information is **literally added** by a **"+"/ plus** operation.

*There are various encoding schemes out there, such as **absolute encoding** or **relative encoding** for example. These seemingly small details can have a significant impact on how the transformer ends up performing. Check out the Swin Transformer paper for some empirical evidence.*

GPT-3 Tensors and the overall Number of Parameters

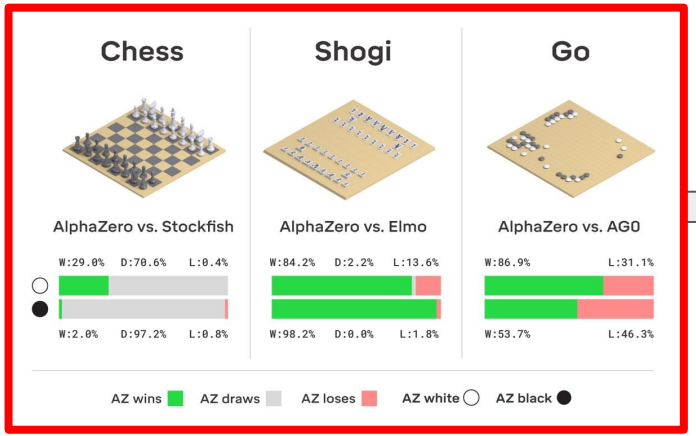
3Blue1Brown

175,181,291,520

Embedding	$d_embed^{12,288} * n_vocab^{50,257}$	$= 617,558,016$
Key	$d_query^{128} * d_embed^{12,288} * n_heads^{96} * n_layers^{96}$	$= 14,495,514,624$
Query	$d_query^{128} * d_embed^{12,288} * n_heads^{96} * n_layers^{96}$	$= 14,495,514,624$
Value	$d_value^{128} * d_embed^{12,288} * n_heads^{96} * n_layers^{96}$	$= 14,495,514,624$
Output	$d_embed^{12,288} * d_value^{128} * n_heads^{96} * n_layers^{96}$	$= 14,495,514,624$
Up-projection	$n_neurons^{49,152} * d_embed^{12,288} * n_layers^{96}$	$= 57,982,058,496$
Down-projection	$d_embed^{12,288} * n_neurons^{49,152} * n_layers^{96}$	$= 57,982,058,496$
Unembedding	$n_vocab^{50,257} * d_embed^{12,288}$	$= 617,558,016$

ChatGPT Pipeline

1) Pretraining the **base model**.
2) Supervised Fine-Tuning (**SFT**).
3) **Reward** Modeling / **RM**.
4) Reinforcement Learning / **RL** (with human in the loop / human feedback).
5) Additional **Chain-of-Thought (CoT)** Reinforcement Learning.



DeepMind subsequently created **AlphaStar AlphaFold, AlphaEvolve** using similar principles.

Conjecture: ChatGPT works so well, because it borrowed many insights from the AlphaZero games playing engine from back in 2016. Words seem to be the new chess pieces that have to be smartly arranged.

Pretraining the Foundational Model

- Use **raw data** to train a **Base Model**.
- The dataset is **huge** => potentially low quality in some places, but **very large quantity** (ex: from roughly 1 to 45 TB / Terabytes of text, and beyond!).
- We obtain a **document completer** in the end, i.e. a very well informed “auto-complete”.
- **Thousands of GPUs** work in parallel (ex: 1000-2000 A100 GPUs).

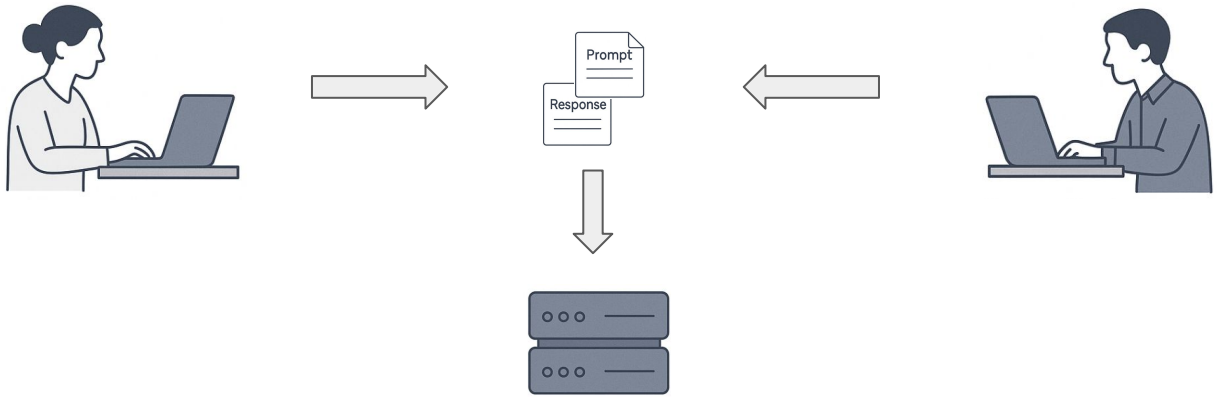


Germany's fastest supercomputer JUWELS at Forschungszentrum Jülich

https://www.fz-juelich.de/en/ias/jsc/systems/supercomputers/**apply-for-computing-time**

Supervised Fine Tuning (SFT stage)

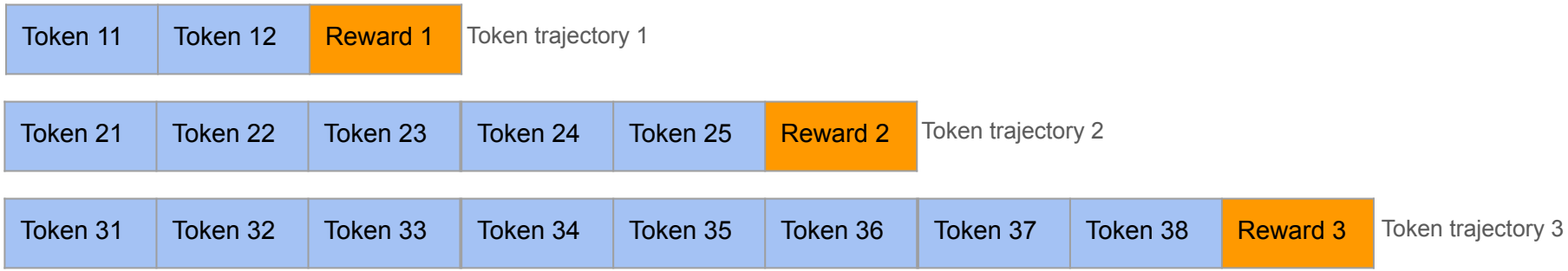
- **Low quantity, but very high quality data**: ~ 100K (prompt, response) tuples.
- Domain Specialists / Contractors have to scrutinize the dataset so that close to ideal **(prompt, response) tuples** are assembled.
- Less GPUs are required than in step 1 (ex: 1-100).
- The outcome is the so-called **SFT model**.



[The figure sub-components were generated using the OpenAI diffusion model]

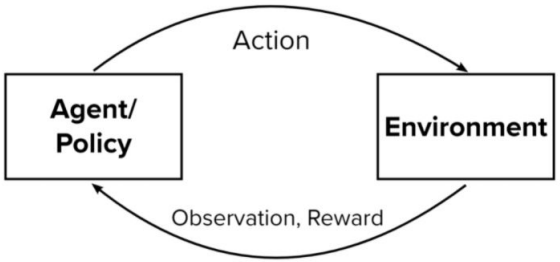
Reward Modeling

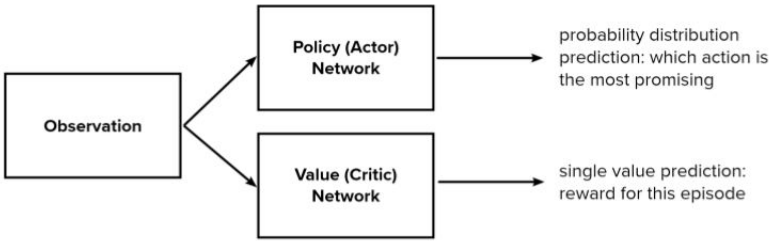
- Ask the SFT model to produce **multiple answers** per prompt.
- Ask human contractors to carefully **rank these answers**.
- Train a reward model on these rankings.
- Requires on the order of 1 to 100 GPUs for training.
- The outcome is the so-called **Reward Model / RM** => **Evaluates token trajectories.**



RLHF Reinforcement Learning

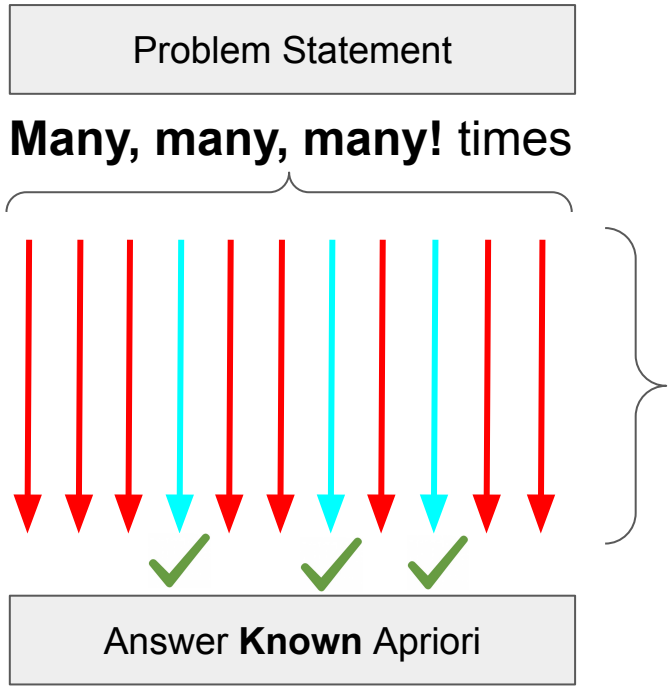
- Train a **Proximal Policy Optimization (PPO)** algorithm.
- Use the previously trained reward model to **evaluate the reward**.
- PPO will have the task of generating **token "trajectories"** that will have a **very good overall score**.
- Requires on the order of 1 to 100 GPUs for training.
- The outcome is the so called **RLHF model** (reinforcement learning with **human feedback**).





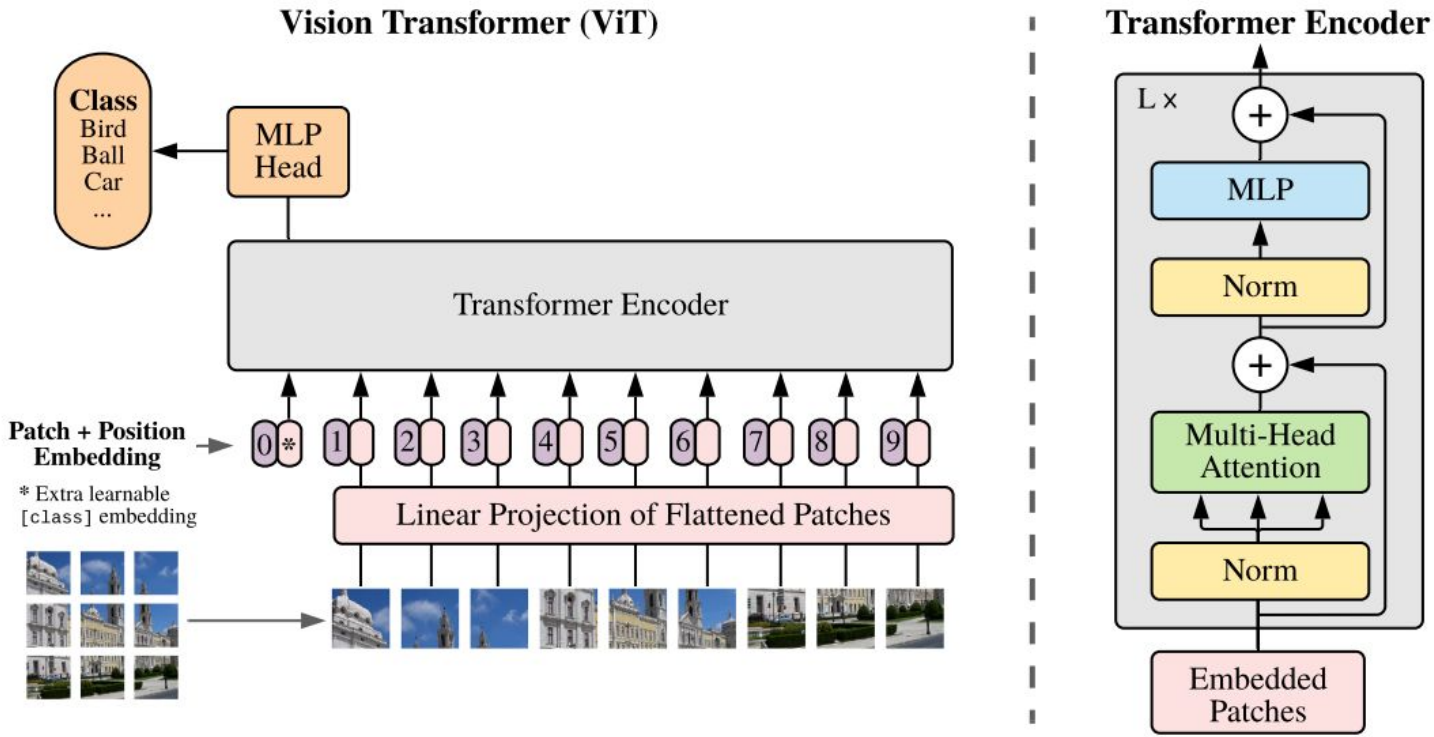
Chain-Of-Thought Reasoning

To generate the recent **chain-of-thought** reasoning systems, we need **even more Reinforcement Learning**, i.e. something more and **different** than RLHF.



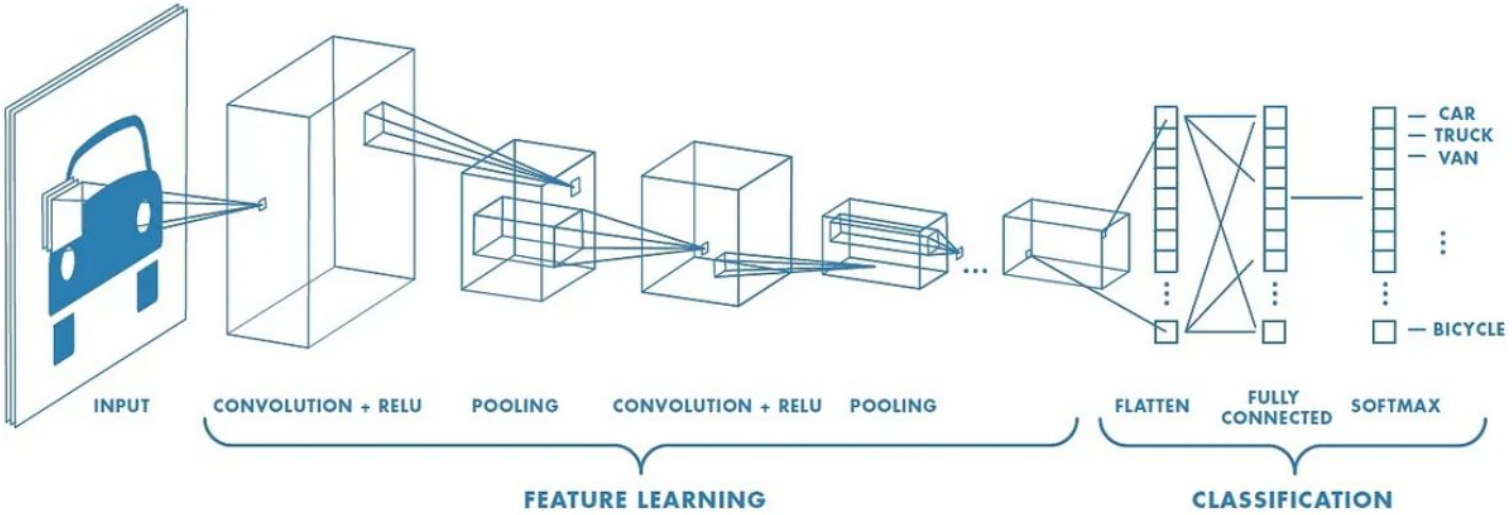
Model **tries** different token trajectories ***on its own*** and **stores the ones that lead to the correct answer**. Then, it ***trains on the successful token trajectories*** to improve its reasoning abilities, such that on future problems, it will **prefer more reasonable chains-of-thought,** i.e. **more "correct thinking and planning"** to solve problems. *One interesting aspect is that training on coding tasks seems to generally improve logic ability across other areas such as math and sciences.*

Vision Transformer / ViT



“An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale” by A. Dosovitskiy et al. (2021)

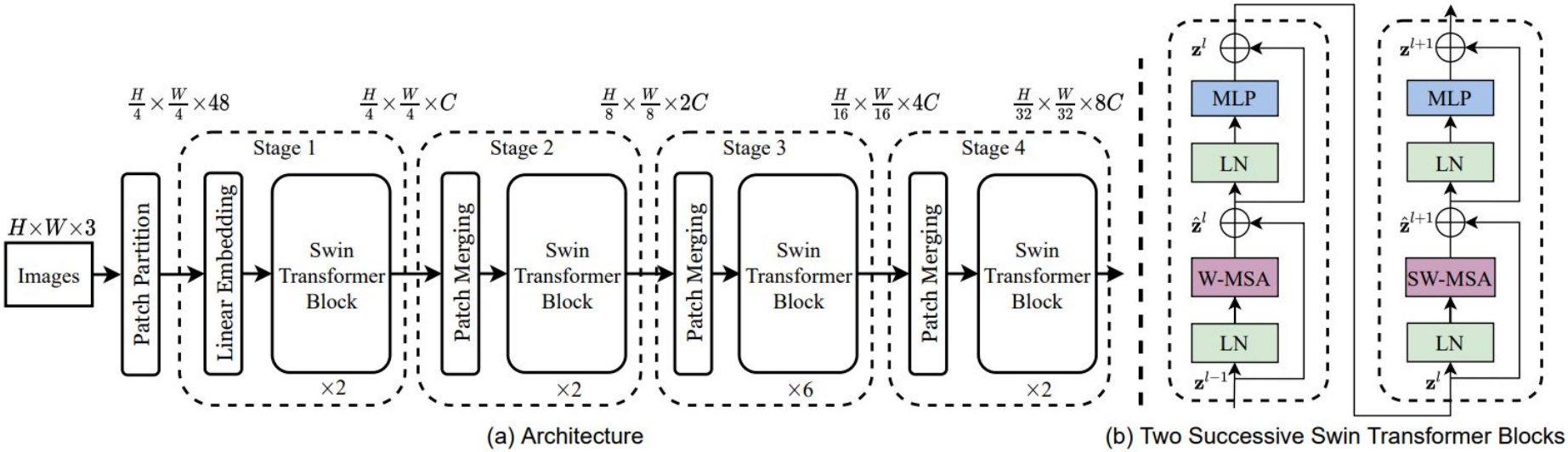
Convolutional Neural Networks (CNNs)



- Generally applied in **computer vision** tasks, to extract **visual features** such as horizontal **edges**, vertical edges, edges of different angles, as well as more **complex shapes** like squares, circles, etc.
- CNNs are very **amenable to parallelism**, so they don't slow down the overall Transformer pipeline.
- **Several Convolutional Layers** are used **at the input** stage of a vision transformer **as an interface** that converts **2D-Maps to 1D-Tokens**, projecting the task to something very similar to the natural language processing scenario.

*Visualize interactive demo at -> **https://cs231n.github.io/convolutional-networks***

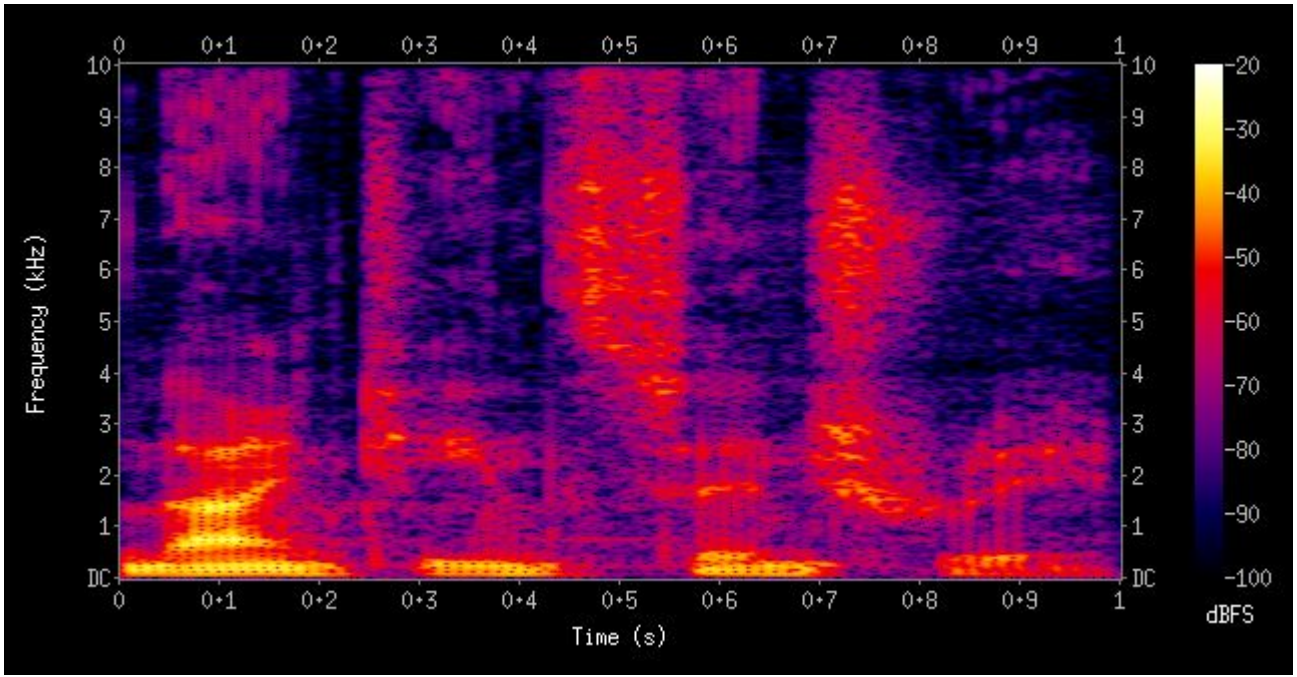
Swin Transformer



“Swin Transformer: Hierarchical Vision Transformer using Shifted Windows” by Liu et al. (2021)

Audio Transformer

Raw sound waves can be mapped to a different space using **STFT** (Short-time Fourier Transform).



Here, time series become images => **problem adapted** to be tackled by a ViT / Swin Transformer.

Time Series Transformer (TST)

Continuous data is **sampled** and **quantized** into discrete tokens.

Implementations available at -> ***https://huggingface.co/docs/transformers/model_doc/time_series_transformer***

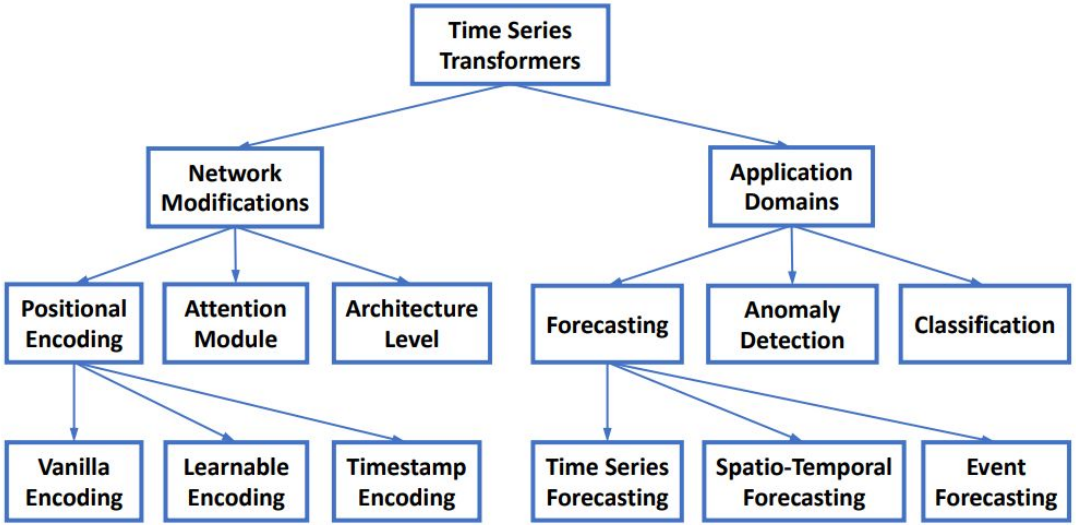
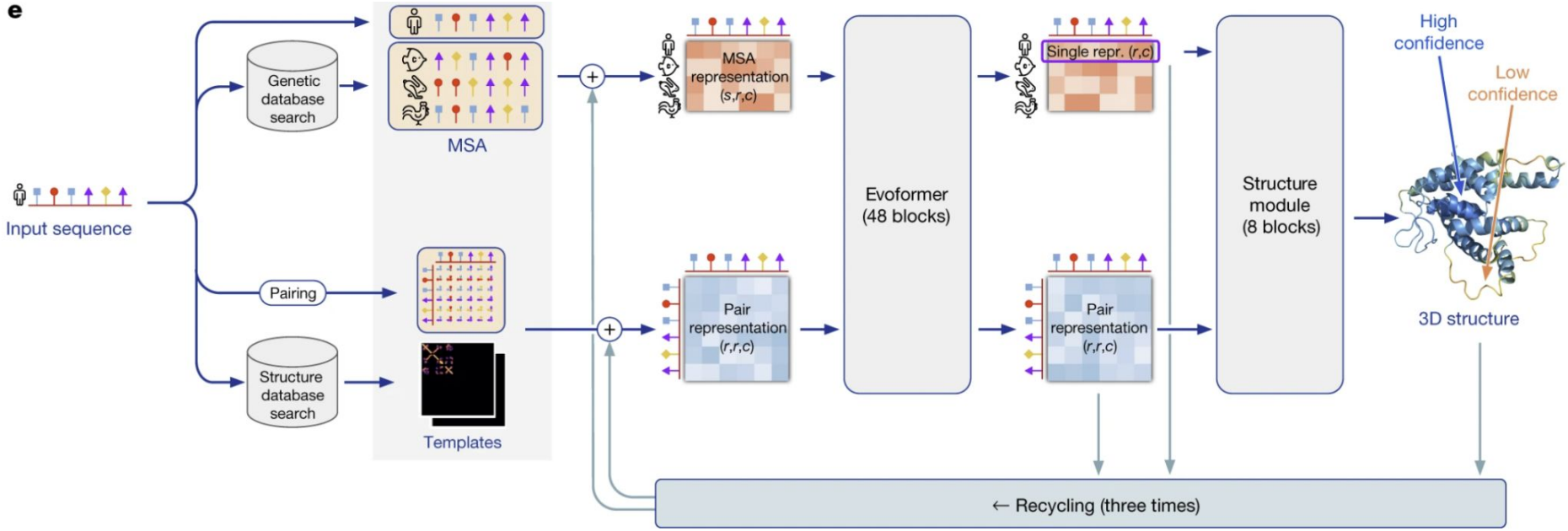


Figure 1: Taxonomy of Transformers for time series modeling from the perspectives of network modifications and application domains.

"Transformers in Time Series: A Survey" paper, by Wen et al. (2023)



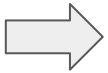
“Highly accurate protein structure prediction with AlphaFold” by John Jumper et al. (Nature, 2021)

- **AlphaFold2** offers many insights on using **Transformers in** relatively **data-scarce regimes**.
- It is also very insightful for cases were you need to **handle 1D, 2D and 3D data formats** within a single pipeline.
- It’s not just a transformer, it’s a hybrid system in conjunction with a **combination of many subtle training tricks** to make protein structure prediction very accurate in the end, acceptable enough for real-world laboratory experiments.

Code example: Self-Attention Snippet Version 1

```
# Version 1:
# We want x[b,t] = mean_{i<=t} x[b,i]
xbow = torch.zeros((B,T,C))
for b in range(B):
    for t in range(T):
        xprev = x[b,:t+1] # (t,C)
        xbow[b,t] = torch.mean(xprev, 0)

print(x[0])
print(xbow[0])
```

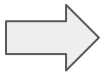


```
torch.Size([4, 8, 2])
tensor([[ 1.9269,  1.4873],
        [ 0.9007, -2.1055],
        [ 0.6784, -1.2345],
        [-0.0431, -1.6047],
        [-0.7521,  1.6487],
        [-0.3925, -1.4036],
        [-0.7279, -0.5594],
        [-0.7688,  0.7624]])
tensor([[ 1.9269,  1.4873],
        [ 1.4138, -0.3091],
        [ 1.1687, -0.6176],
        [ 0.8657, -0.8644],
        [ 0.5422, -0.3617],
        [ 0.3864, -0.5354],
        [ 0.2272, -0.5388],
        [ 0.1027, -0.3762]])
```

Code example: Self-Attention Snippet Version 2

```
# Version 1
# We want x[b,t] = mean_{i<=t} x[b,i]
xbow = torch.zeros((B,T,C))
for b in range(B):
    for t in range(T):
        xprev = x[b,:t+1] # (t,C)
        xbow[b,t] = torch.mean(xprev, 0)
print(x[0])
print(xbow[0])

# Version 2
wei = torch.tril(torch.ones(T, T))
wei = wei / wei.sum(1, keepdim=True)
xbow2 = wei @ x # (B, T, T) @ (B, T, C) --->
(B, T, C)
print("Are xbow and xbow2 the same? -> ",
torch.allclose(xbow, xbow2))
```

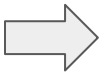


```
torch.Size([4, 8, 2])
tensor([[ 1.9269,  1.4873],
        [ 0.9007, -2.1055],
        [ 0.6784, -1.2345],
        [-0.0431, -1.6047],
        [-0.7521,  1.6487],
        [-0.3925, -1.4036],
        [-0.7279, -0.5594],
        [-0.7688,  0.7624]])
tensor([[ 1.9269,  1.4873],
        [ 1.4138, -0.3091],
        [ 1.1687, -0.6176],
        [ 0.8657, -0.8644],
        [ 0.5422, -0.3617],
        [ 0.3864, -0.5354],
        [ 0.2272, -0.5388],
        [ 0.1027, -0.3762]])
Are xbow and xbow2 the same? ->  True
```

Code example: Self-Attention Snippet Version 3

```
# Version 2
wei = torch.tril(torch.ones(T, T))
wei = wei / wei.sum(1, keepdim=True)
xbow2 = wei @ x # (B, T, T) @ (B, T, C) --->
(B, T, C)
print("Are xbow and xbow2 the same? -> ",

# Version 3: using Softmax
tril = torch.tril(torch.ones(T,T))
wei = torch.zeros((T,T))
wei = wei.masked_fill(tril == 0,
float('-inf'))
wei = F.softmax(wei, dim=-1)
xbow3 = wei @ x
print("Are xbow/xbow2 equal to xbow3? -> ",
torch.allclose(xbow, xbow3))
```



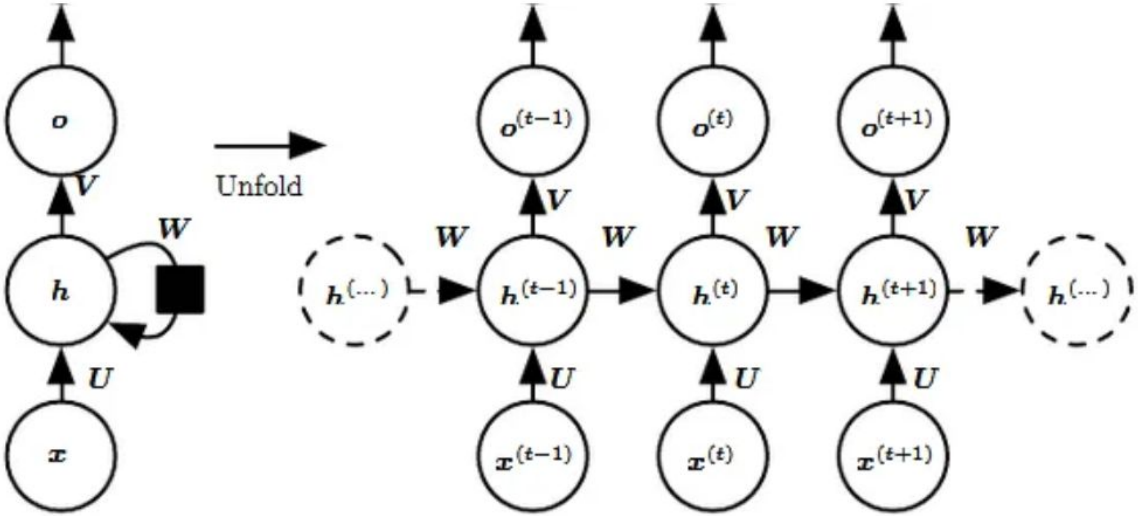
```
torch.Size([4, 8, 2])
tensor([[ 1.9269,  1.4873],
        [ 0.9007, -2.1055],
        [ 0.6784, -1.2345],
        [-0.0431, -1.6047],
        [-0.7521,  1.6487],
        [-0.3925, -1.4036],
        [-0.7279, -0.5594],
        [-0.7688,  0.7624]])
tensor([[ 1.9269,  1.4873],
        [ 1.4138, -0.3091],
        [ 1.1687, -0.6176],
        [ 0.8657, -0.8644],
        [ 0.5422, -0.3617],
        [ 0.3864, -0.5354],
        [ 0.2272, -0.5388],
        [ 0.1027, -0.3762]])
Are xbow/xbow2 equal to xbow3? ->  True
```

Takeaways

- Generative AI is the **art of encoding complex real world distributions**, such that we can **generate creative results** later **via sampling** from the learned distribution.
- Transformers are powerful neural networks that **borrow the best ideas** from prior models in the AI ecosystem and **combine them together for a synergistic effect**.
- **Self-attention** and **Feed-Forward MLP** are the major conceptual components of a **Transformer block**.
- **Self-attention** is essentially a **communication graph** where tokens exchange **information stored in channels** among themselves.
- The **Feed-Forward MLP** is the memory database of the transformer, used to enrich the input embeddings with new facts.
- **Residual additive connections** and **pre- / post-normalization** are important details / attributes, that contribute towards a more successful training and faster convergence.

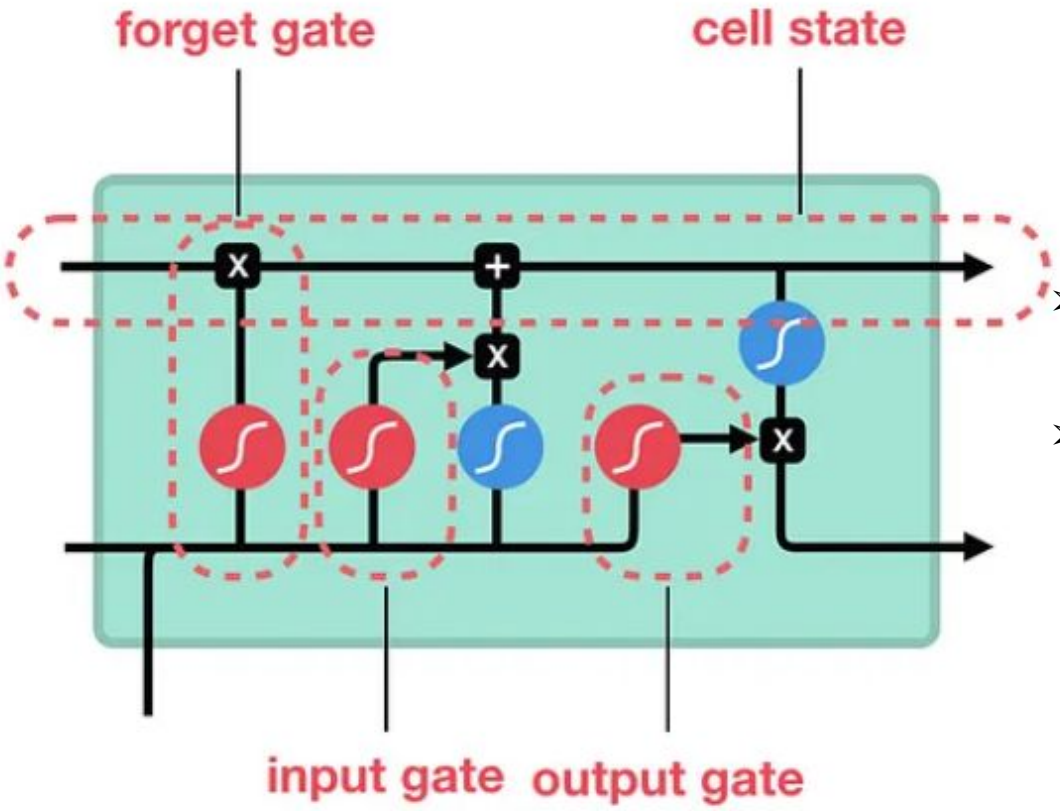
Bonus slides and helper material next...

Sequential nature of Recurrent Neural Networks (RNNs)



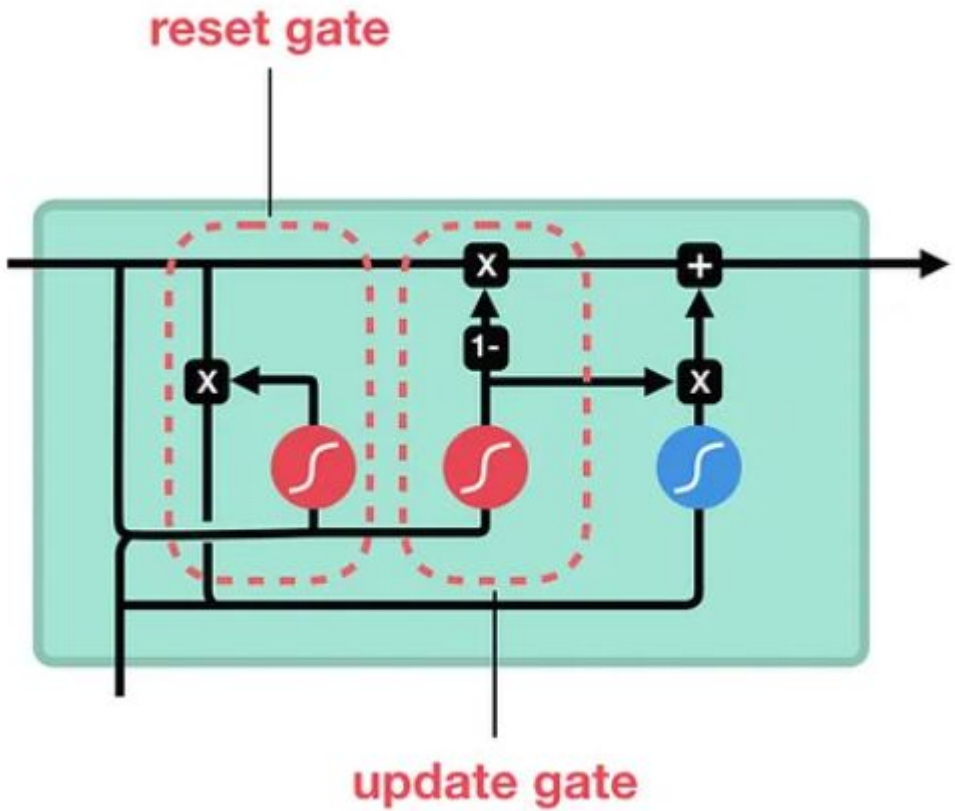
- By **unfolding** the feedback loop in time, we see the actual complexity of these networks. It is as if we train a **very deep network** and that may be very well why they're significantly harder to train.
- With **RNNs** things happen **sequentially** => **deep** graph structure.
- With **Transformers** things happen **in parallel** => **broad** graph structure.
- **Transformers** might simply be **easier to train stably**, and maybe that's why they have better overall results.

Long short-term memory (LSTMs)



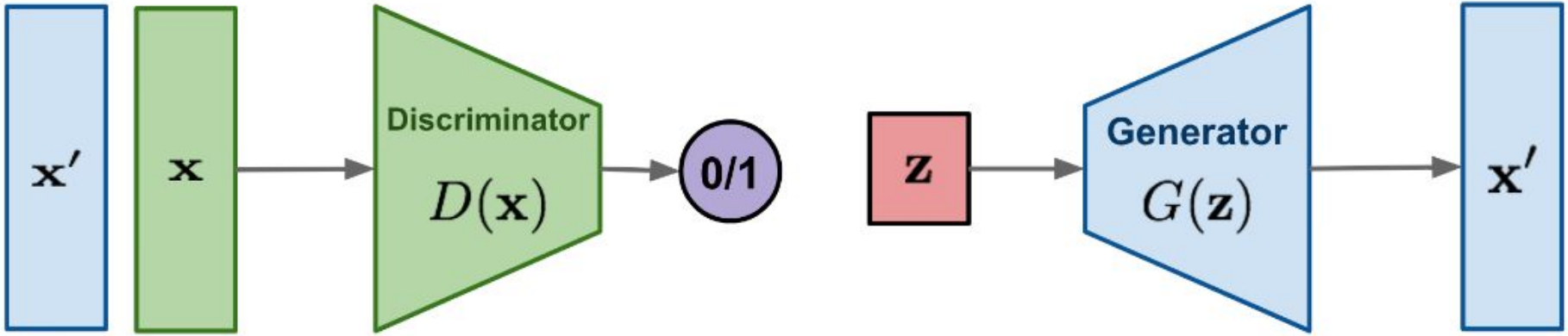
- Contains special **gates** that address the problem of **vanishing gradients**.
- Addresses the problem of **exploding gradients**.

Gated Recurrent Units (GRUs)



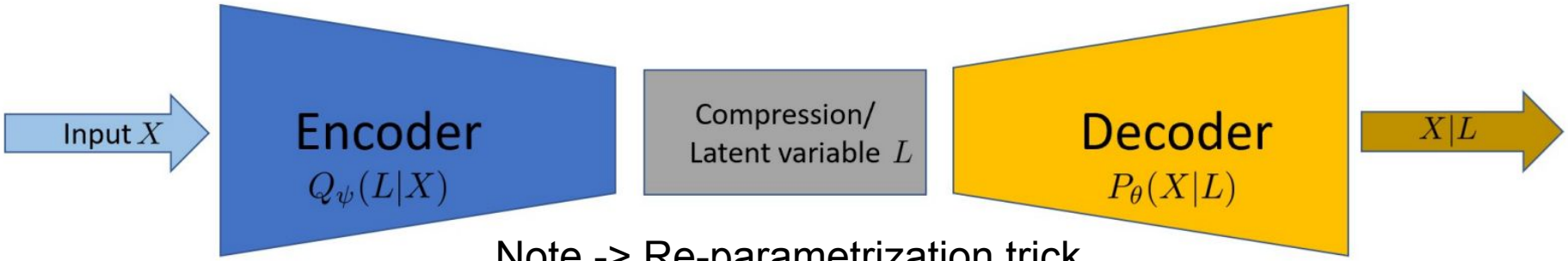
- Generally considered **faster** than LSTMs.
- In practice => **similar outcomes to** what **LSTM** models provide.

Generative Adversarial Networks / GANs



- Two networks: Generator and Discriminator play a **min-max game**
- **Generator** aims at producing realistic outputs **to trick the Discriminator**
- **Discriminator** strives to improve its ability to **discern true from fake** data

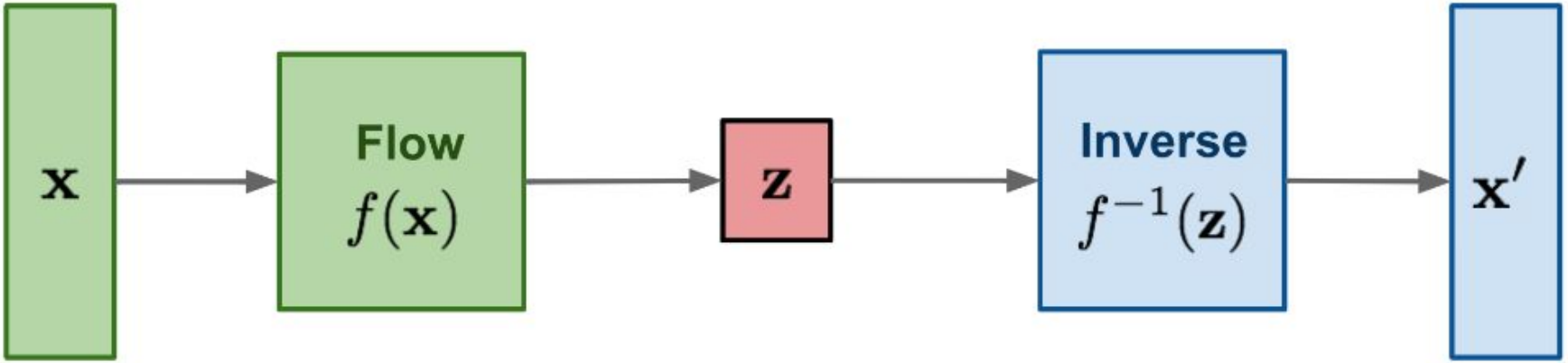
Variational Autoencoders / VAEs



Note -> Re-parametrization trick

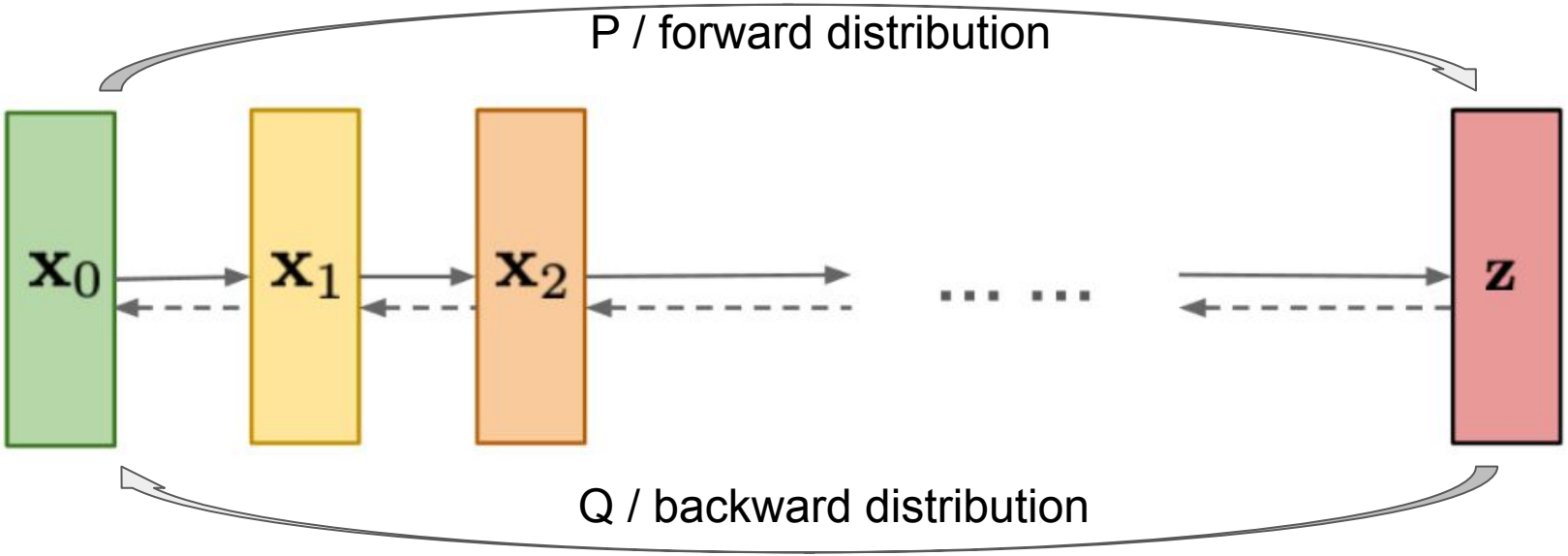
- Inspired from traditional **autoencoders** that **compress** data into **vector codes**
- Variational autoencoders will learn a **latent distribution function** instead
- The **latent distribution can be sampled**, resulting in variable latent samples
- The samples are **decoded**, resulting in a **variety of realistic reconstructions**

Flow-based models



- Use **invertible neural networks** to learn the data distribution
- Enable **exact log-likelihood** directly thanks to **one-to-one** invertibility
- **More** computationally **expensive than VAE**s, due to invertibility requirements

Diffusion models



- **Markov chain** of states, where each state is a noisy version of the other
- The original input X_0 is **diffused into** pure **noise Z** over several steps
- The model **learns the noise that needs to be removed from a state** at time **t**, to get to an **earlier state**

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- “Attention is all you need” by Vaswani et al. -> https://arxiv.org/pdf/1706.03762.pdf
- LLM visualization tool -> https://bbycroft.net/llm
- 3Blue1Brown channel / playlist -> https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
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